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Configuring a single and redundant XF204BA DNA Y switch with an S7-1500H

SCALANCE XF204BA DNA, SIMATIC S7-1500H

<https://support.industry.siemens.com/cs/ww/en/view/109816704>

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1 Introduction

1.1 Overview

Redundant S7-1500H system

The redundant S7-1500H system features high reliability and plant availability. A redundant configuration of critical automation components reduces the likelihood of production stops and the consequences of component faults.

With firmware version V3.0 or below, a redundant system requires a ring topology with S2/S1 devices; the H CPUs must be part of this ring.

New feature/function

As of firmware version V3.0 and TIA Portal V18, the redundant S7-1500H system also supports PROFINET R1 system redundancy. In addition, H CPUs no longer have to be a part of a PROFINET ring (also known as an MRP ring) or an MRP domain. New topology variants are possible because of this, such as a linear topology, mixed topology, or a configuration with no additional devices.

Note

For more information on system redundancy terminology, see [chapter 4.1](#).

Motivation and content of this application

The new features let you construct the redundant S7-1500H system in differing variants.

This application example will consider the following three scenarios:

1. Setting up an S7-1500H with R1 devices and a Y switch in a linear topology
2. Setting up an S7-1500H with R1 devices in a ring topology and a Y switch with a lower-level linear topology
3. Setting up an S7-1500H with R1 devices and Y switches with DNA redundancy in a ring topology

All three scenarios will focus on the following three points:

- Explanations of the structure (theory)
- Indications that must be followed during configuration
- A treatment of the switchover/failure behavior in case of a failure of a controller, IO device or network

[Chapter 2](#) goes into greater depth on the third scenario with a hands-on commissioning example and step-by-step configuration instructions in TIA Portal.

1.2 Relevant modules

1.2.1 Y switch and DNA

If you wish to use both R1 devices as well as S2/S1 devices in your setup, then your S2/S1 device connection will need a Y switch with the Dual Network Access function. It connects S2/S1 devices arranged in an PROFINET ring with an R1 system (redundant configuration).

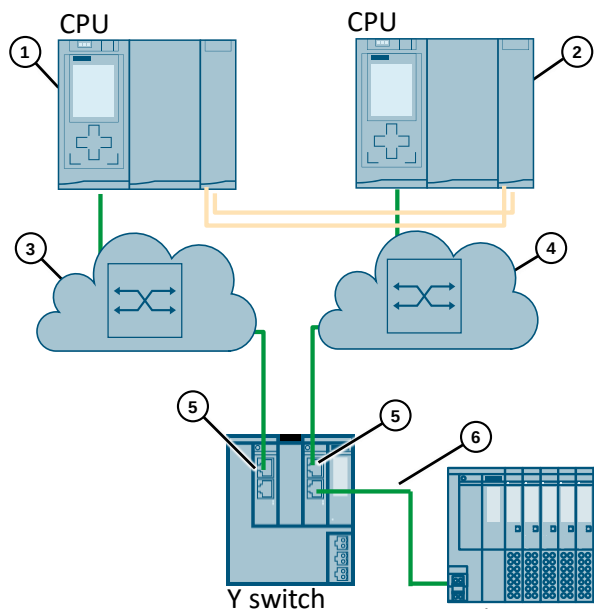
Dual Network Access (DNA)

DNA is a technology for connecting one network with two networks that are decoupled from one another. A switch that provides this feature is known as a "Y switch".

The ports on the Y switch for connecting the two decoupled networks are the DNA ports. The Y switch ensures that the two networks remain isolated from each other.

The following Figure shows how a Y switch is employed:

Figure 1-1



The numbers have the meanings as follows:

Table 1-1

No.	Meaning	Note
1.	First redundant controller	
2.	Second redundant controller	
3.	Network of the first controller	The topology is variable.
4.	Network of the second controller	The topology is variable.
5.	DNA ports of the Y switch	Devices in network ③ cannot access the devices in network ④, and vice versa.
6.	Shared network with S1/S2 devices (here: linear topology)	The remaining ports on the Y switch that are not DNA ports connect the two networks ③ and ④ with this network. The devices in this network are accessible from network ③ as well as from network ④.

Dual Network Access redundancy (DNA redundancy)

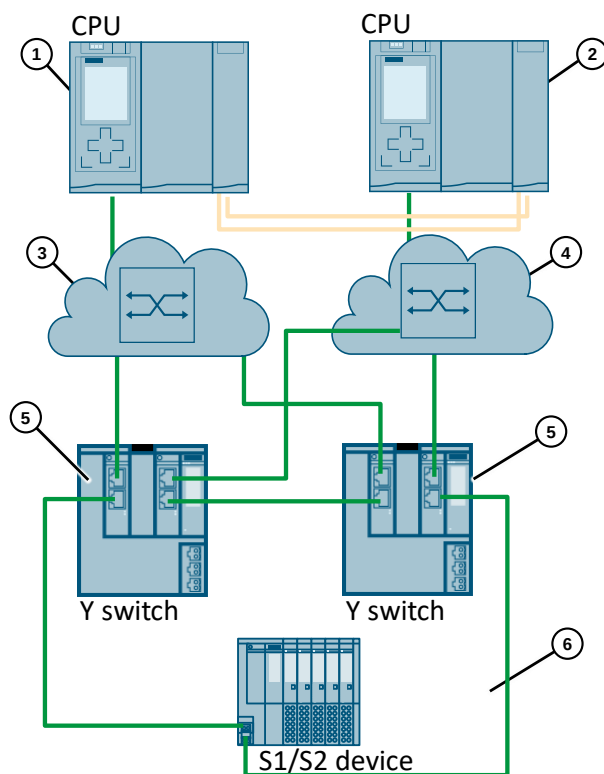
To increase the availability of the network, you can also implement the Y switch with a redundant design (DNA redundancy). Three Y switches are used for this. DNA redundancy strictly requires a PROFINET ring in the shared network.

One Y switch takes on the roles of MRP manager and DNA manager. A second Y switch functions as MRP client and DNA client. The connection to the two decoupled networks is enabled if at least one Y switch is in operation.

In regular operation, the DNA ports of the DNA client are blocked; the DNA ports of the DNA manager are in the "Forwarding" state.

The Figure below shows you how a redundant Y switch is employed:

Figure 1-2



The numbers have the meanings as follows:

Table 1-2

No.	Meaning	Note
1.	First redundant controller	
2.	Second redundant controller	
3.	Network of the first controller	The topology is variable.
4.	Network of the second controller	The topology is variable.
5.	Redundant Y switches	Every Y switch is connected with network ③ and ④ via the switch's DNA ports. One Y switch is the DNA manager. The other Y switch is the DNA client and blocks its DNA ports. If the DNA manager fails, the DNA client opens its DNA ports.
6.	Common PROFINET ring with S1/S2 devices	The Y switch with the "DNA manager" role is automatically the "MRP manager". The Y switch with the "DNA client" role is automatically the "MRP client".

SCALANCE XF204-2BA DNA

You can use the SCALANCE XF204-2BA DNA as a Y switch.

Figure 1-3



The switch has the following features:

- The Y switch interfaces S2/S1 devices with a fault-tolerant R1 system.
- The switch supports the DNA manager and DNA client function.
- The Y switch feature allows the system to maintain communication in a PROFINET ring within a redundant system.

Note

More information on the Y switch can be found in the module manual (see \4\ in [chapter 5.3](#)).

Note

The SCALANCE XF204-2BA DNA will be referred to as the "Y switch" in the rest of this application example.

1.2.2 ET 200SP with R1 functionality

For redundant applications in the field of distributed I/O devices, the ET 200M for PROFIBUS and the ET 200SP HA for PROFINET have been available up to now. These two series are complemented by the ET 200SP for PROFINET and can be configured on the fault-tolerant SIMATIC S7-1500H(F) as of TIA Portal V18.

Description

The redundant ET 200SP R1 system guarantees high availability thanks to redundant PROFINET connections. Unlike S2 devices, R1 devices are equipped with two interface modules. Should one interface module fail, the H CPUs can still reach the R1 device via the second interface module. This grants R1 devices higher availability than S2 devices.

Figure 1-4



Sample configuration of a plant with an ET 200SP R1

To set up an ET 200SP R1 station, you will need at least the following components:

- SIMATIC system rail (qty 1)
- BaseUnit BU type M0 (qty 1)
- Interface module IM 155-6 PN R1 (qty 2)
- Identical SIMATIC BusAdapters (qty 2)
- BaseUnits and electronics modules
- Server module (qty 1)

Note

More information on the SIMATIC ET 200SP R1 can be found in the module manual (see \7\ in [chapter 5.3](#)).

1.3 Components used

This application example was created with these hardware and software components:

Table 1-3

Device	Components	Quantity	Item number	Note
2nd Y switch	SCALANCE XF204-2BA	1	6GK5204-2BA00-2YF2	Pre-assembled with 2x BA 2xRJ45 HA bus adapters V4.3
1st Y switch	SCALANCE XF204-2BA	1	6GK5204-2AA00-2YF2	V4.3
	BA 2xRJ45	2	6ES7193-6AR00-0AA0	Bus adapter for Y switch
1st/2nd switch	SCALANCE XB208	2	6GK5208-0BA00-2AB2	V4.2
ET 200SP(S2)	IM 155-6 PN/2 HF	1	6ES7155-6AU01-0CN0	V4.2
	BaseUnit A0 dark	2	6ES7193-6BP00-0BA0	
	BaseUnit A0 light	1	6ES7193-6BP20-0DA0	
	BA 2xRJ45	1	6ES7193-6AR00-0AA0	You can also use other BAs from the ET 200SP portfolio. Both BAs must be identical.
	DI8x24VDC	1	6ES7131-6BF00-0CA0	V2.0
	DO8x24VDC/0.5A ST	2	6ES7132-6BF00-0BA0	V1.1
	Server module	1	6ES7193-6PA00-0AA0	V1.1
ET 200SP(R1)	IM 155-6 PN R1	4	6ES7155-6AU00-0HM0	V6.0
	BaseUnit M0	2	6ES7193-6BR00-0HM0	
	BaseUnit A0 dark	1	6ES7193-6BP00-0BA0	
	BaseUnit A0 light	1	6ES7193-6BP20-0DA0	
	BA 2xRJ45	4	6ES7193-6AR00-0AA0	You can also use other BAs from the ET 200SP portfolio. Both BAs must be identical.
	DO8x24VDC/0.5A ST	1	6ES7132-6BF01-0BA0	V1.1
	AI 2xU/I 2-/4-wire HF	1	6ES7134-6HB00-0CA1	V2.0
	Server module	2	6ES7193-6PA00-0AA0	V1.1
H CPU	CPU S7-1517H-3PN (bundle)	1	6ES7500-0HP00-0AB0	V3.0 S7-1500H bundle consisting of 2 1517H-3 PN CPUs, four synchronization modules (up to 10 m), and two 1-meter synchronization cables.
	SIMATIC S7 Memory Card	2	6ES7954-8LL03-0AA0	SIMATIC S7 Memory Card, 256 MB You can also use a different storage capacity.

Device	Components	Quantity	Item number	Note
Programming device	Programming device with TIA Portal V18			
All	Power supplies, power cables and PROFINET cables are assumed to be on hand.			

The components listed here can be obtained from the [Siemens Industry Mall](#).

This application example consists of the following components:

Table 1-4

Components	File name	Note
Document	109816704_S7-1500H_with_Y-Switch_V1_0_en.pdf	
Project	109816704_Redundant_DNA.zip	TIA Portal V18

2 Configuration examples with R1 system redundancy

2.1 General configuration information

If you set up and configure the redundant S7-1500 system with R1 system redundancy, then follow the indications below.

Note

Drag all devices that you will use in the redundant system into the Network view of TIA Portal and configure the topology. This will grant you the best diagnostic options in the event of a fault.

2.1.1 Setup rules for the topologies

MRP roles in the PROFINET ring of the H CPU

If you use a PROFINET ring as a topology, then the MRP roles are defined as follows:

- H CPU: "MRP-Manager (auto)"
- IO devices: "MRP-Client"

MRP roles when using a Y switch with lower-level PROFINET ring

If you are using a Y switch with a lower-level PROFINET ring to link to S1/S2 devices, then the MRP roles are defined as follows:

- For the Y switch: "MRP-Manager"
- Both Y switches with DNA redundancy:
 - One Y switch: "MRP-Manager"
 - One Y switch: "MRP-Client"
- For the IO devices: "MRP-Client"

DNA roles with DNA redundancy

If you are using redundant Y switches with DNA redundancy, then the following assignment applies automatically:

- The Y switch with the "MRP-Manager" role is simultaneously the "DNA manager".
- The Y switch with the "MRP-Client" role is simultaneously the "DNA client".

Note

This assignment cannot be changed.

MRP ports and DNA ports

When configuring MRP in TIA Portal, you can define the MRP ports for every device.

With the Y switch, you can only define the MRP ports, but not the DNA ports. Ports that are not configured as MRP ports are thus automatically the DNA ports.

Maximum number of IO devices

The following applies for the maximum number of PROFINET devices:

- In the PROFINET ring: 50
- IO devices connectable to the H CPUs: 256

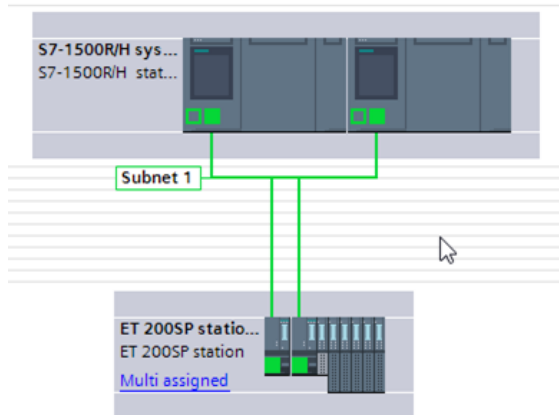
2.1.2 PROFINET subnet and IO devices

Shared PROFINET subnet

The controllers and IO devices are connected in TIA Portal with a shared PN subnet. This configuration enables a two-way connection of S1 or S2 devices using a Y switch.

The screenshot below shows the configuration in TIA Portal:

Figure 2-1



Response watchdog time

The reconfiguration time of the PROFINET rings is typically less than 200 milliseconds. In order for the data exchange to work after a reconfiguration, you must set the response watchdog time of the IO devices that are nodes in a PROFINET ring to be greater than 200 milliseconds.

If the response watchdog time of the nodes is too low, TIA Portal will throw an error message about the response watchdog time during compilation. Set the response watchdog time as specified in the error message.

Note

You can use the S7-1500R/H add-in to correctly set the response watchdog time in a redundant system. You can find the add-in under reference \8\ in [chapter 5.3](#).

SCALANCE devices

You have the option of configuring the SCALANCE devices via their Web Based Management feature (abbreviated as WBM), or in TIA Portal.

It is recommended to integrate the SCALANCE devices within the redundant system as IO devices. This allows you to analyze the PROFINET diagnostics from the SCALANCE devices and troubleshoot rapidly.

2.1.3 Assignment

Multi assigned

To assign IO devices with R1 system redundancy to the S7-1500H redundant system, connect every interface module of the R1 device in TIA Portal with an H CPU. The left-hand interface module of each R1 device must be connected with the left H CPU depicted in the Network view. The right-hand interface module of each R1 device must be connected with the right H CPU depicted in the Network view.

To assign IO devices with S2 system redundancy to the S7-1500H redundant system, connect the interface module of the IO device in the TIA Portal Network view with each H CPU.

Standard IO devices with S1 system redundancy can be connected to the redundant system with the "Switched S1 device" function. Configure this function by assigning a standard IO device to both H CPUs in the redundant system.

Note

Even if the S2 or S1 device is only connected to one H CPU, assign the IO device to both H CPUs in TIA Portal. Otherwise, you will receive an error message when compiling.

Note

TIA Portal will show "Multi assigned" for the IO devices.

Change operating mode

If an IO device with S2 system redundancy is only connected with one H CPU, for example, a switch in a linear topology, then the preset "IO device(S2)" operating mode will trigger the "MAINT" diagnostic message to be sent to the H CPUs, and the MAINT LEDs of the H CPU will light up.

To set this correctly, change the "Operating mode" parameter of the IO device from "IO device(S2)" to "IO device(S1)". You can find the setting for the "Operating mode" parameter in TIA Portal in the "I/O communication" view for each S2 device.

Note

To get to the "I/O communication" view, go to the TIA Portal Network view and select the redundant system. In the table view of the Network view, change to "I/O communication".

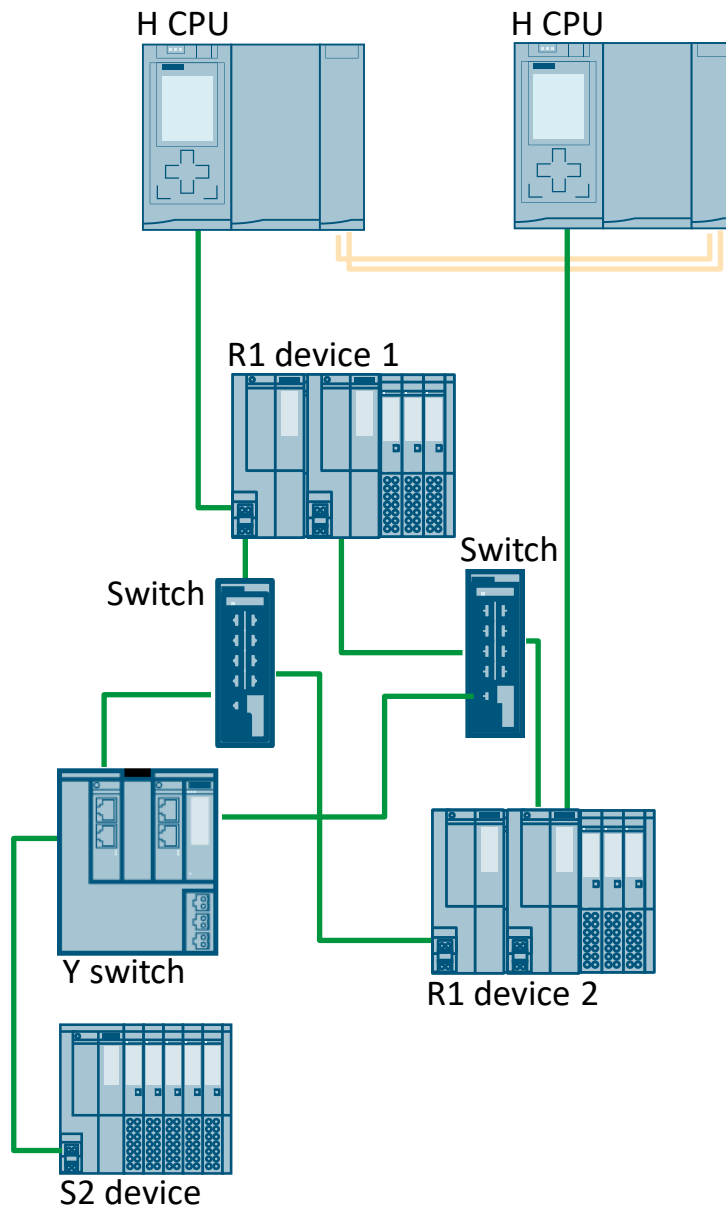
2.2 Linear topology with R1 devices and a Y switch

2.2.1 Structure and description

Structure

The following section shows you the abstract structure of the redundant S7-1500H system with R1 devices and a Y switch in a linear topology:

Figure 2-2



Description

Communication in the redundant S7-1500 system runs on two separate PROFINET lines. The "R1 device 1" and "R1 device 2" devices are connected directly to the PROFINET line 1 and PROFINET line 2. The PROFINET lines feed into the devices "R1 device 1" and "R1 device 2" devices from multiple directions. This increases availability in case of a double line break.

An additional switch is needed for each PROFINET line. The Y switch is connected with one of these switches via each of its DNA ports.

The S2 device is connected to one of the ports on the Y switch that is not a DNA port. The S2 device can be reached both from PROFINET line 1 as well as from PROFINET line 2.

Note

Since the Y switch only has two DNA ports (one DNA port per H CPU), the line cannot be extended onward. An additional switch is needed for each PROFINET line.

You can also connect the Y switch as the last element in the PROFINET line. In this case, you do not need the two additional switches.

Note

To increase availability, you can also run a PROFINET ring on the ports of the Y switch that are not DNA ports.

Configuration notes

Note the following pointers for configuration in TIA Portal:

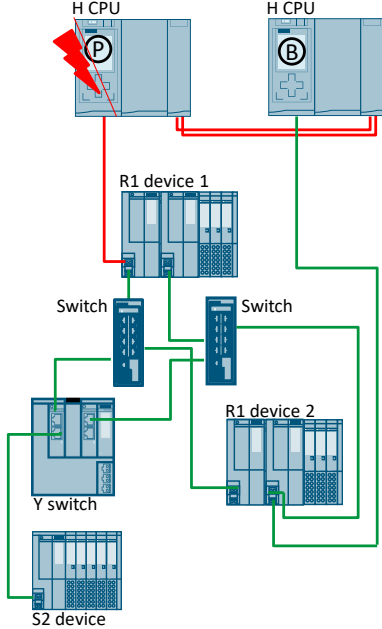
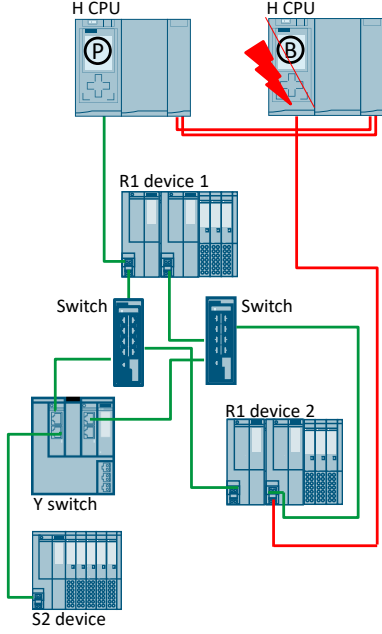
- A PROFINET ring is not needed. The MRP roles are set to "Not device in the ring" for all devices.
- There can be up to 256 devices in the redundant system.
- The two additional switches are each connected to only one H CPU. Configure the switches in TIA Portal and assign the switches to both H CPUs anyway (Multi assigned).
- Change the operating mode of the two switches to "IO device(S1)".
- The topology below the Y switch can vary. If you are using a PROFINET ring, then you will need an MRP domain. In this case, the Y switch takes on the MRP role "MRP manager" and the S2 device takes on the MRP role "MRP client".

2.2.2 Response to errors

Failure of an H CPU

The Table shows the redundancy scenarios in the event that an H CPU fails:

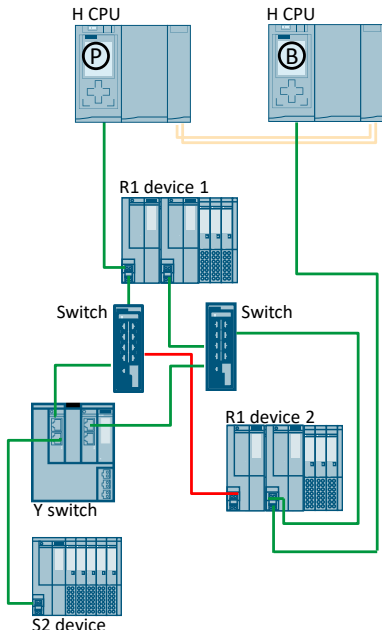
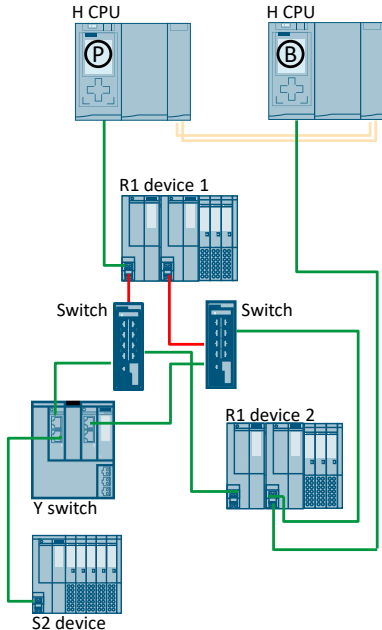
Table 2-1

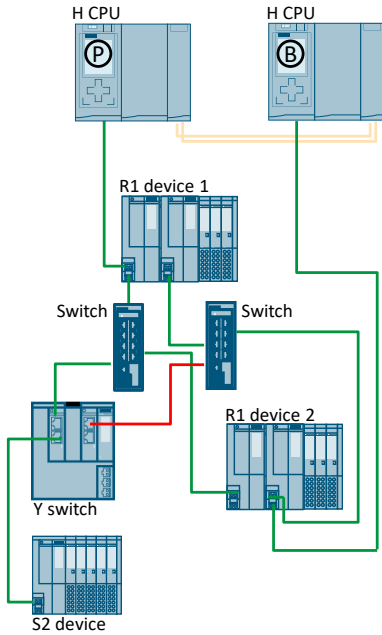
Scenario	Description
 The diagram shows a redundant system with two H CPUs at the top. The left H CPU is marked with a red lightning bolt and a circled 'P', indicating it has failed. The right H CPU is marked with a circled 'B', indicating it is the backup. Below the CPUs are two R1 devices (R1 device 1 and R1 device 2), two switches, a Y switch, and an S2 device. Red lines represent the primary CPU's connections, and green lines represent the backup CPU's connections. The backup CPU (B) is shown with active connections to all devices, while the primary CPU (P) has a lightning bolt over it, indicating failure.	The primary(P) CPU has failed. The redundant system switches over to the backup CPU. The backup CPU becomes the new primary CPU and switches to the system state RUN Solo. The redundant system thereby reaches all IO devices in the linear topology again. The redundancy of the system is limited. This limitation has no impact on the process. If another system component or another network segment fails, then the redundant S7-1500H system can fail. The IO devices are not longer fully reachable in this case.
 The diagram shows the same redundant system as above. In this scenario, the right H CPU (the backup) is marked with a red lightning bolt and a circled 'B', indicating it has failed. The left H CPU is marked with a circled 'P', indicating it is the primary. The primary CPU (P) maintains active connections (green lines) to all devices (R1 device 1, R1 device 2, switches, Y switch, and S2 device). The backup CPU (B) has a lightning bolt over it, indicating failure.	The backup (B) CPU has failed. The redundant system switches to the system state RUN Solo. A switchover is not necessary. All IO devices in the linear topology remain reachable. The redundancy of the system is limited. This limitation has no impact on the process. If another system component or another network segment fails, then the redundant S7-1500H system can fail. The IO devices are not longer fully reachable in this case.

Line break

The Table shows possible redundancy scenarios in the event of a line break:

Table 2-2

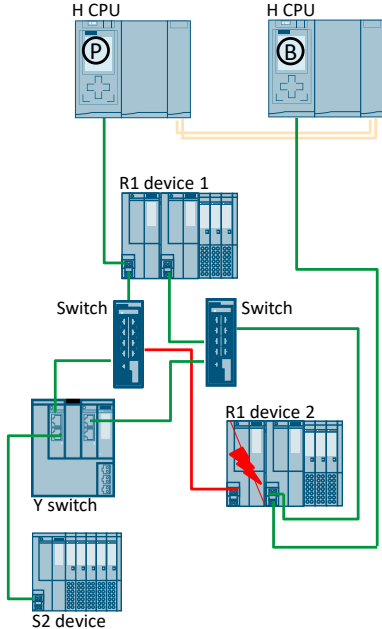
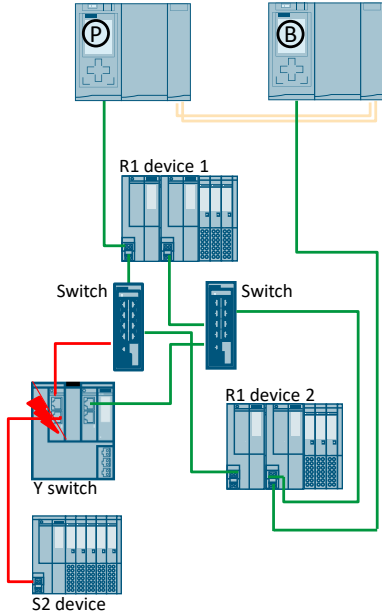
Scenario	Description
	The PROFINET line between the switch and the right-hand interface module of the "R1 device 2" has failed. The primary CPU and backup CPU remain in the RUN Redundant mode. The redundant system selects an alternative connection to the device "R1 device 2" via the backup CPU. The redundant system thereby reaches all IO devices in the linear topology again. The redundancy of the system is limited. This limitation has no impact on the process. If another system component or another network segment fails, then the redundant S7-1500H system can fail. The IO devices are not longer fully reachable in this case.
	Both PROFINET lines between the two interface modules of the device "R1 device 1" and the two switches have failed (double failure scenario). The primary CPU and backup CPU remain in the RUN Redundant mode. With the PROFINET line is fed from multiple sources, the redundant system selects an alternative connection to the Y switch and the device "R1 device 2": • The device "R1 device 1" remains reachable via the primary CPU. • The device "R1 device 2" and the Y switch are reachable via the backup CPU. The redundant system thereby reaches all IO devices in the linear topology again. The redundancy of the system is limited. This limitation has no impact on the process. If another system component or another network segment fails, then the redundant S7-1500H system can fail. The IO devices are not longer fully reachable in this case.

Scenario	Description
 The diagram illustrates a redundant S7-1500H system configuration. At the top, two H-CPU units are shown: the primary CPU (P) on the left and the backup CPU (B) on the right. They are connected to two R1 devices, R1 device 1 and R1 device 2. R1 device 1 is connected to a switch on the left, and R1 device 2 is connected to a switch on the right. Both switches are connected to a Y-switch at the bottom left and an S2 device at the bottom right. A red line connects the left switch to the Y-switch, indicating a failed connection. The Y-switch is also connected to the S2 device. The S2 device is connected to the backup CPU (B) via a green line. The primary CPU (P) is connected to R1 device 1 via a green line, and R1 device 1 is connected to the left switch via a green line. The backup CPU (B) is connected to R1 device 2 via a green line, and R1 device 2 is connected to the right switch via a green line. The right switch is connected to the Y-switch via a green line. The Y-switch is connected to the S2 device via a green line. The S2 device is connected to the backup CPU (B) via a green line. The primary CPU (P) is connected to the backup CPU (B) via an orange line.	The PROFINET line between the switch and the right-hand interface module of the Y switch has failed. The primary CPU and backup CPU remain in the RUN Redundant mode. The redundant system requires no alternative connection, since the primary CPU can still reach all IO devices. The redundancy of the system is limited. This limitation has no impact on the process. If another system component or another network segment fails, then the redundant S7-1500H system can fail. The IO devices are not longer fully reachable in this case.

Failure of one interface module

The Table shows possible redundancy scenarios should one interface module fail:

Table 2-3

Scenario	Description
 The diagram illustrates a network topology with two H CPUs (Primary and Backup) connected to R1 device 1 and R1 device 2. R1 device 1 is connected to a Switch, which is connected to a Y switch. The Y switch is connected to an S2 device. R1 device 2 is also connected to the Switch. A red arrow points to the left-hand interface module (slot 0) of R1 device 2, indicating a failure.	The left-hand interface module (slot 0) of the device "R1 device 2" fails. The roles of the primary and backup CPU do not change. The redundant system selects an alternative connection to the device "R1 device 2" via the backup CPU. The redundant system thereby reaches all IO devices in the linear topology again. The redundancy of the system is limited. This limitation has no impact on the process. If another system component or another network segment fails, then the redundant S7-1500H system can fail. The IO devices are not longer fully reachable in this case.
 The diagram illustrates a network topology with two H CPUs (Primary and Backup) connected to R1 device 1 and R1 device 2. R1 device 1 is connected to a Switch, which is connected to a Y switch. The Y switch is connected to an S2 device. R1 device 2 is also connected to the Switch. A red arrow points to the left-hand interface module (slot 0) of the Y switch, indicating a failure.	The left-hand interface module (slot 0) of the Y switch fails. The roles of the primary and backup CPU do not change. The redundant system selects an alternative connection to the Y switch via the backup CPU. Due to the linear topology in the lower-level PROFINET network, the device "S2 device" is no longer reachable. The redundancy of the system is limited. To reduce the probability of failure of the device "S2 device", connect the device "S2 device" to the Y switch with a PROFINET ring.

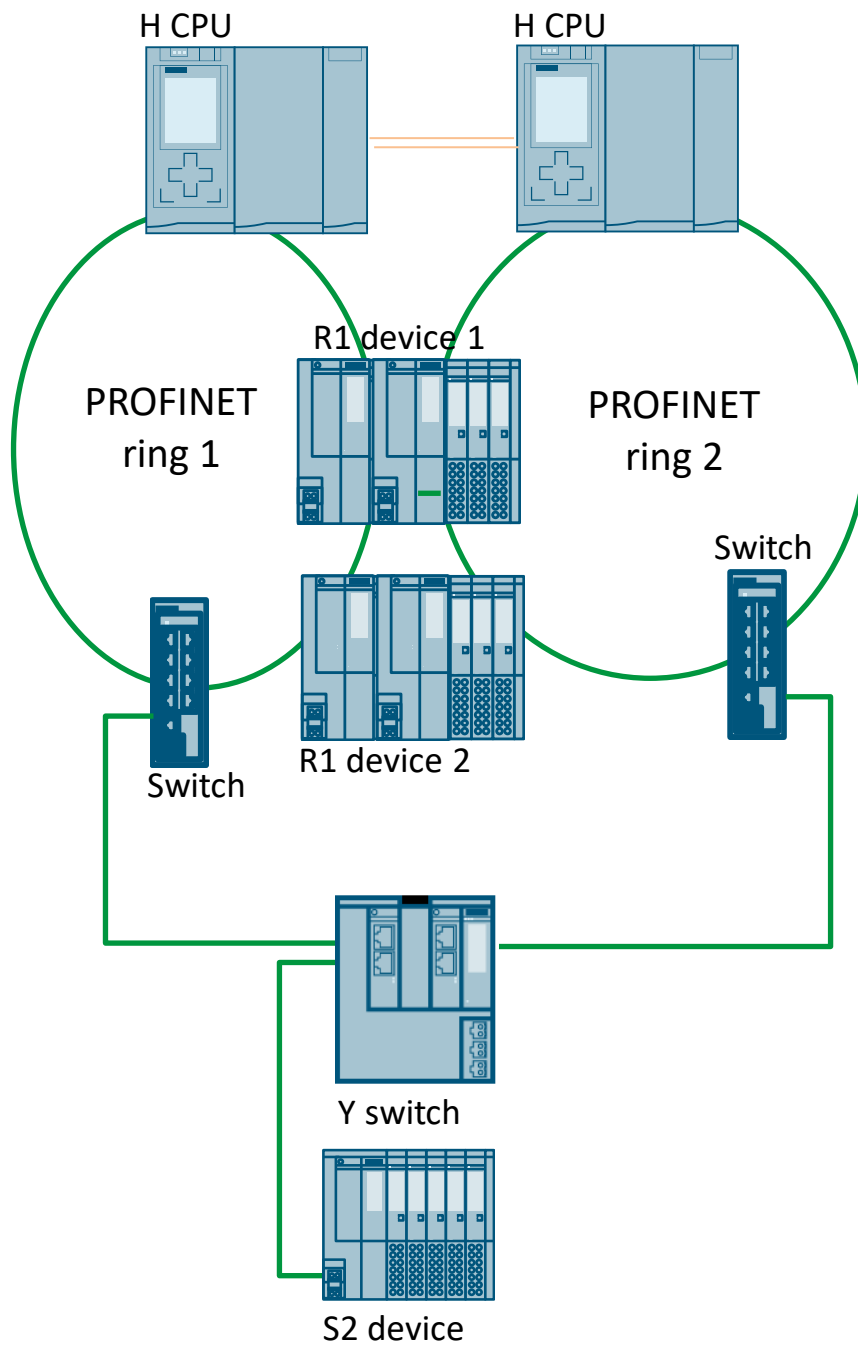
2.3 Ring topology with R1 devices and a Y switch

2.3.1 Structure and description

Structure

The following section shows the abstract structure of the redundant S7-1500H system with R1 devices and a Y switch in a ring topology:

Figure 2-3



Description

Communication in the redundant S7-1500 system runs on two separate PROFINET rings. The devices "R1 device 1" and "R1 device 2" are directly connected to both PROFINET rings. An additional switch is required for each PROFINET ring. The DNA ports of the Y switch are each connected via one of these switches with PROFINET ring 1 or PROFINET ring 2 of the redundant S7-1500H system. The S2 device is connected to one of the ports on the Y switch that is not a DNA port. The S2 device can be reached both from PROFINET ring 1 as well as from PROFINET ring 2.

Note

To increase availability, you can also run a PROFINET ring on the ports of the Y switch that are not DNA ports.

Configuration notes

Note the following pointers for configuration in TIA Portal:

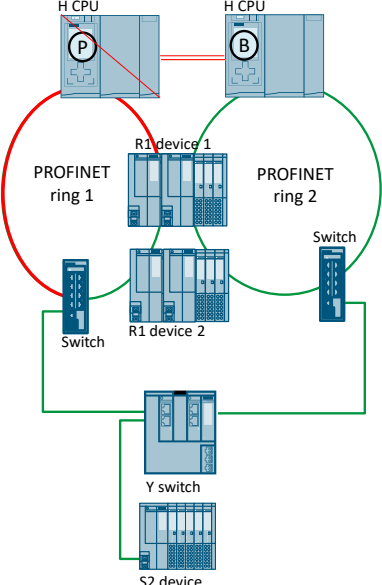
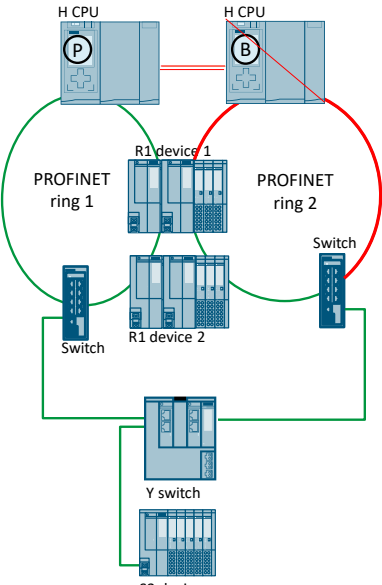
- Two separate PROFINET rings are used. Create two MRP domains in TIA Portal for this. The H CPUs assume the MRP role of "MRP manager" while the IO devices assume the MRP role of "MRP client".
- The maximum number of PROFINET devices in the PROFINET ring of the fault-tolerant system is 50 nodes.
- The two additional switches are each connected to only one H CPU. Configure the switches in TIA Portal and assign the switches to both H CPUs anyway ("Switched S1 device").
- Change the operating mode of the two switches to "IO device(S1)".
- The topology below the Y switch can vary. If you are using a PROFINET ring here as well, then you will need a third MRP domain. In this case, the Y switch takes on the MRP role "MRP manager" and the S2 device takes on the MRP role "MRP client".

2.3.2 Response to errors

Failure of an H CPU

The Table shows the redundancy scenarios in the event that an H CPU fails:

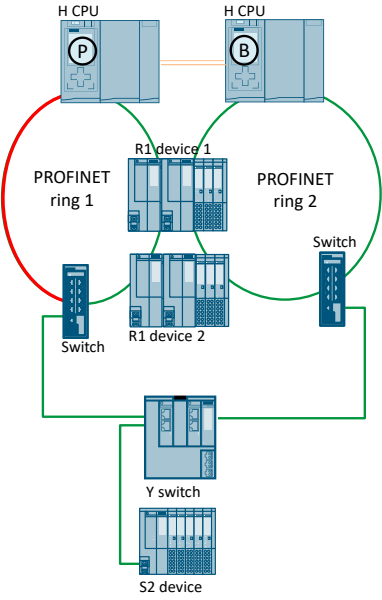
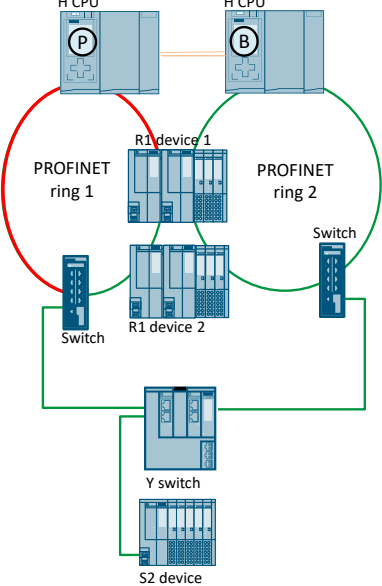
Table 2-4

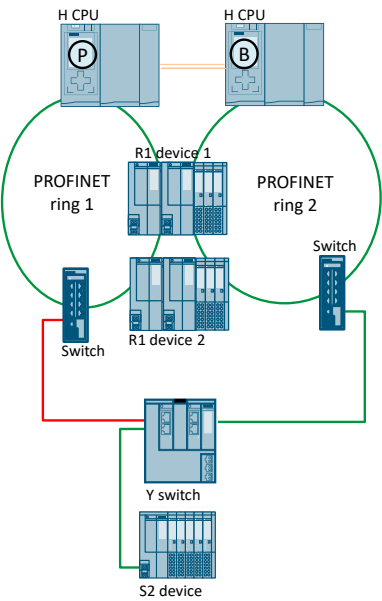
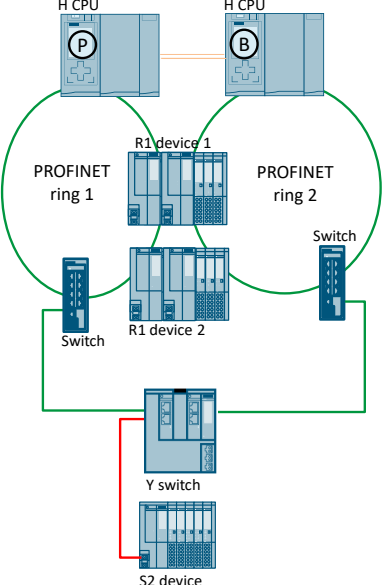
Scenario	Description
	The primary (P) CPU has failed. The redundant system switches over to the backup CPU. The backup CPU becomes the new primary CPU and switches to the system state RUN Solo. The redundant system thereby reaches all IO devices in the PROFINET network again. The redundancy of the system is limited. This limitation has no impact on the process. If another system component or another network segment fails, then the redundant S7-1500H system can fail. The IO devices are not longer fully reachable in this case.
	The backup (B) CPU has failed. The redundant system switches to the system state RUN Solo. A switchover is not necessary. All IO devices remain reachable. The redundancy of the system is limited. This limitation has no impact on the process. If another system component or another network segment fails, then the redundant S7-1500H system can fail. The IO devices are not longer fully reachable in this case.

Line break

The Table shows possible redundancy scenarios in the event of a line break:

Table 2-5

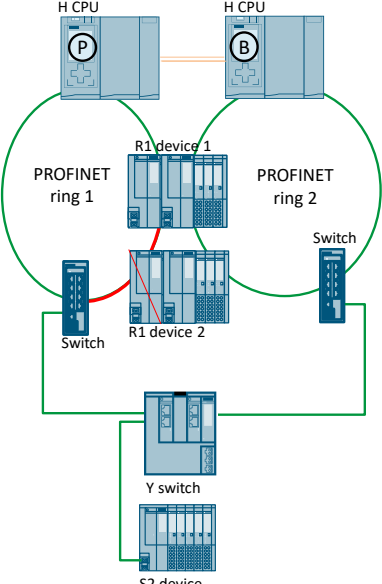
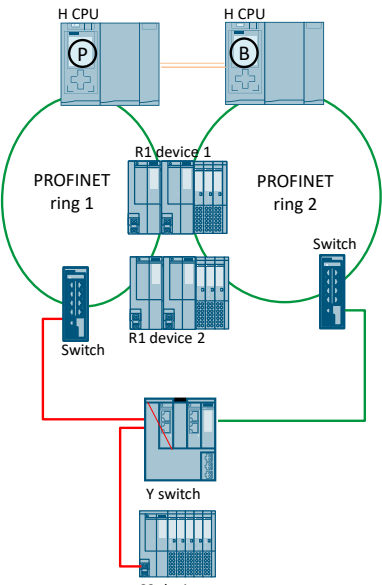
Scenario	Description
	The PROFINET line in PROFINET ring 1 between the primary CPU and the switch has failed. The primary CPU and backup CPU remain in the RUN Redundant mode. The primary CPU can still reach all IO devices in the PROFINET network. The redundancy of the system is limited. This limitation has no impact on the process. If another system component or another network segment fails, then the redundant S7-1500H system can fail. The IO devices are not longer fully reachable in this case.
	Two PROFINET lines in PROFINET ring 1 have failed. The primary CPU and backup CPU remain in the RUN Redundant mode. The redundant system selects an alternative connection to the IO devices via the backup CPU. The redundant system thereby reaches all IO devices in the PROFINET network again. The redundancy of the system is limited. This limitation has no impact on the process. If another system component or another network segment fails, then the redundant S7-1500H system can fail. The IO devices are not longer fully reachable in this case.

Scenario	Description
	The PROFINET line between the switch and the left-hand interface module of the Y switch has failed. The primary CPU and backup CPU remain in the RUN Redundant mode. The roles of the primary and backup CPU do not change. The redundant system selects an alternative connection to the Y switch via the backup CPU. The redundant system thereby reaches all IO devices in the PROFINET network again. The redundancy of the system is limited. This limitation has no impact on the process. If another system component or another network segment fails, then the redundant S7-1500H system can fail. The IO devices are not longer fully reachable in this case.
	The PROFINET line between the left-hand interface module of the Y switch and the device "S2 device" has failed. The roles of the primary and backup CPU do not change. The device "S2 device" is no longer reachable, since there is no alternative path through the linear topology. To reduce the probability of failure of the device "S2 device", connect the device "S2 device" to the Y switch with a PROFINET ring.

Failure of one interface module

The Table shows possible redundancy scenarios should one interface module fail:

Table 2-6

Scenario	Description
	The left-hand interface module (slot 0) of the device "R1 device 2" fails. The roles of the primary and backup CPU do not change. The redundant system selects an alternative connection to the device "R1 device 2" via the backup CPU. The redundant system thereby reaches all IO devices in the PROFINET network again. The redundancy of the system is limited. This limitation has no impact on the process. If another system component or another network segment fails, then the redundant S7-1500H system can fail. The IO devices are not longer fully reachable in this case.
	The left-hand interface module (slot 0) of the Y switch fails. The roles of the primary and backup CPU do not change. The redundant system selects an alternative connection to the Y switch via the backup CPU. The device "S2 device" is no longer reachable, since there is no alternative path through the linear topology. To reduce the probability of failure of the device "S2 device", connect the device "S2 device" to the Y switch with a PROFINET ring.

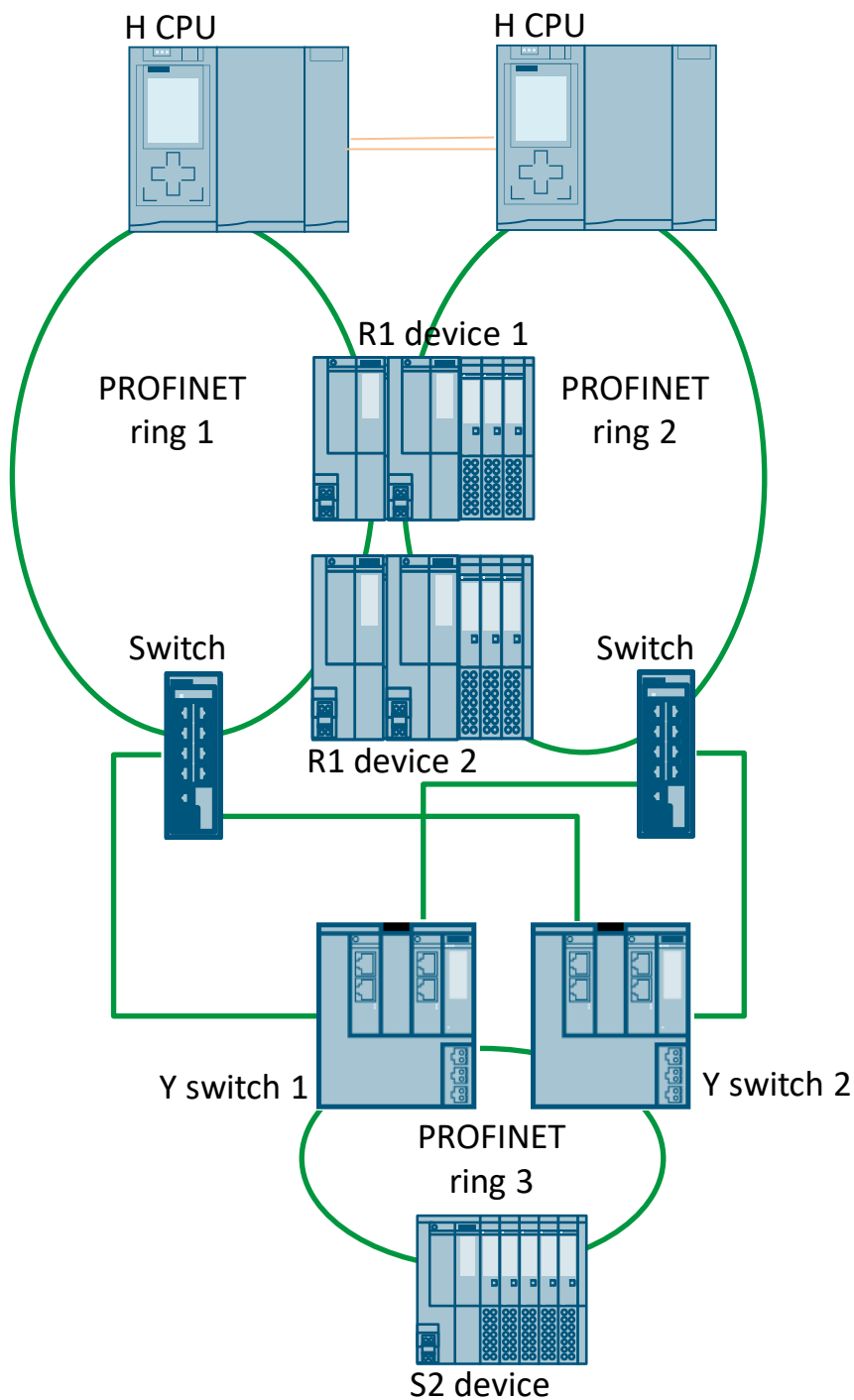
2.4 Ring topology with an R1 device and a redundant Y switch

2.4.1 Structure and description

Structure

The following section shows the structure of the redundant S7-1500H system with R1 devices and a redundant Y switch in a ring topology:

Figure 2-4



Description

Communication in the redundant S7-1500H system runs on two separate PROFINET rings. Two Y switches with DNA redundancy are employed for increased availability.

The devices "R1 device 1" and "R1 device 2" are directly connected to both PROFINET rings.

An additional switch is required for each PROFINET ring. The DNA ports of the two Y switches are each connected via one of these switches with PROFINET ring 1 or PROFINET ring 2 of the redundant S7-1500H system.

The other ports of the Y switches that are not DNA ports, along with the S2 device, are connected with each other via a PROFINET ring (PROFINET ring 3).

The S2 device can be reached both from PROFINET ring 1 as well as from PROFINET ring 2.

Note

If you are using DNA redundancy, then a PROFINET ring in the lower-level network is required.

Configuration notes

Note the following pointers for configuration in TIA Portal:

- Two separate PROFINET rings are used. The H CPUs assume the MRP role of "MRP manager" while the IO devices assume the MRP role of "MRP client".
- The maximum number of PROFINET devices in the PROFINET ring of the fault-tolerant system is 50 nodes.
- The two additional switches are each connected to only one H CPU. Configure the switches in TIA Portal and assign the switches to both H CPUs anyway ("Switched S1 device").
- Change the operating mode of the two switches to "IO device(S1)".
- DNA redundancy requires a PROFINET ring in the lower-level network. This means you need a third MRP domain. A Y switch simultaneously assumes the roles of "MRP manager" and "DNA manager", while the other Y switch takes on the roles of "MRP client" and "DNA client".

Note

[Chapter 3](#) contains instructions on how to set up and configure this scenario in TIA Portal.

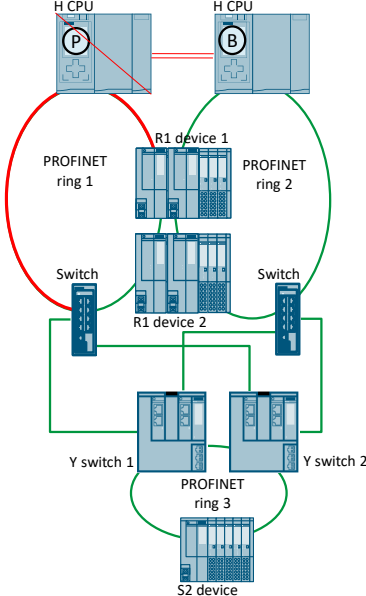
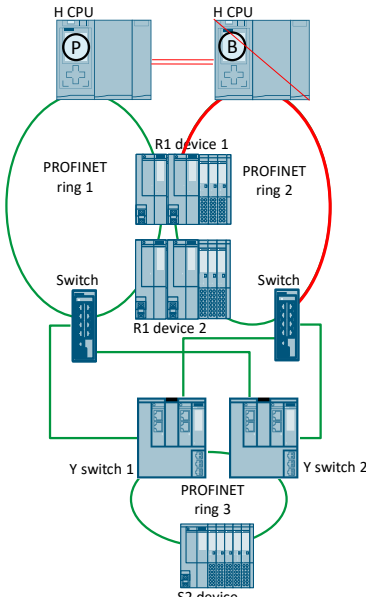
On the HTML page of this application example, you will find a prefabricated TIA Portal project for download.

2.4.2 Response to errors

Failure of an H CPU

The Table shows the redundancy scenarios in the event that an H CPU fails:

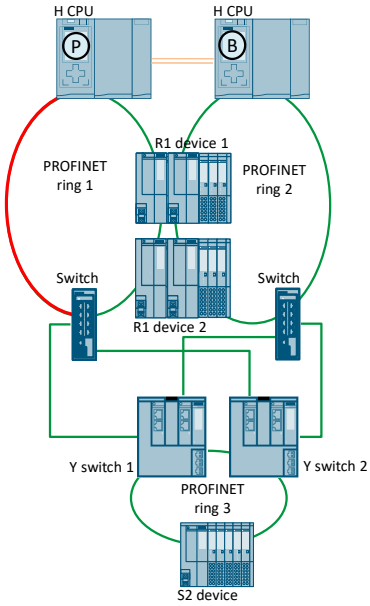
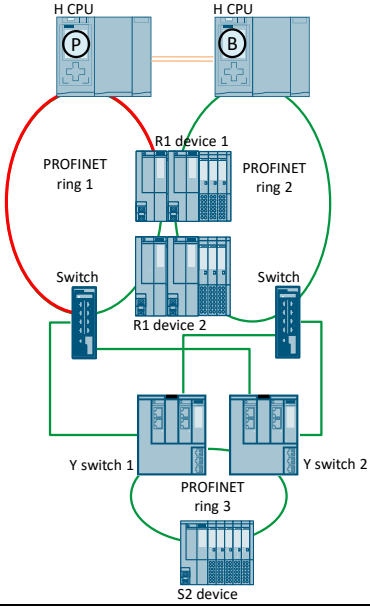
Table 2-7

Scenario	Description
	The primary (P) CPU has failed. The redundant system switches over to the backup CPU. The backup CPU becomes the new primary CPU and switches to the system state RUN Solo. The redundant system thereby reaches all IO devices in the PROFINET network again. The redundancy of the system is limited. This limitation has no impact on the process. If another system component or another network segment fails, then the redundant S7-1500H system can fail. The IO devices are not longer fully reachable in this case.
	The backup (B) CPU has failed. The redundant system switches to the system state RUN Solo. A switchover is not necessary. All IO devices remain reachable. The redundancy of the system is limited. This limitation has no impact on the process. If another system component or another network segment fails, then the redundant S7-1500H system can fail. The IO devices are not longer fully reachable in this case.

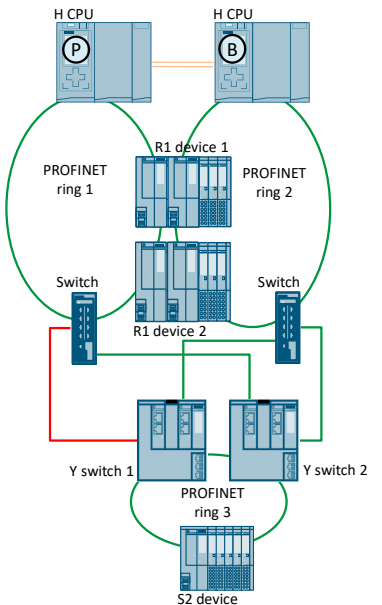
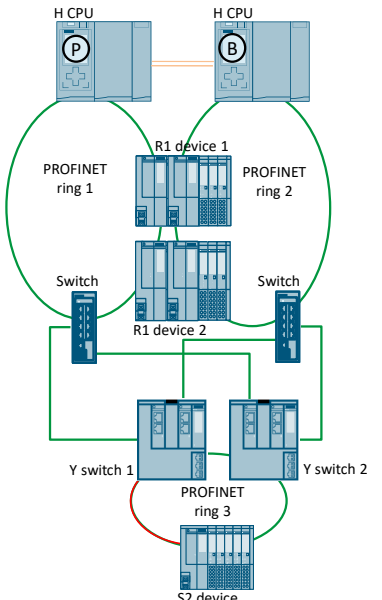
Line break

The Table shows possible redundancy scenarios in the event of a line break:

Table 2-8

Scenario	Description
	The PROFINET line in PROFINET ring 1 between the primary CPU and the switch has failed. The primary CPU and backup CPU remain in the RUN Redundant mode. The primary CPU can still reach all IO devices in the PROFINET network. The redundancy of the system is limited. This limitation has no impact on the process. If another system component or another network segment fails, then the redundant S7-1500H system can fail. The IO devices are not longer fully reachable in this case.
	Two PROFINET lines in PROFINET ring 1 have failed. The primary CPU and backup CPU remain in the RUN Redundant mode. The redundant system selects an alternative connection to the IO devices via the backup CPU. The redundant system thereby reaches all IO devices in the PROFINET network again. The redundancy of the system is limited. This limitation has no impact on the process. If another system component or another network segment fails, then the redundant S7-1500H system can fail. The IO devices are not longer fully reachable in this case.

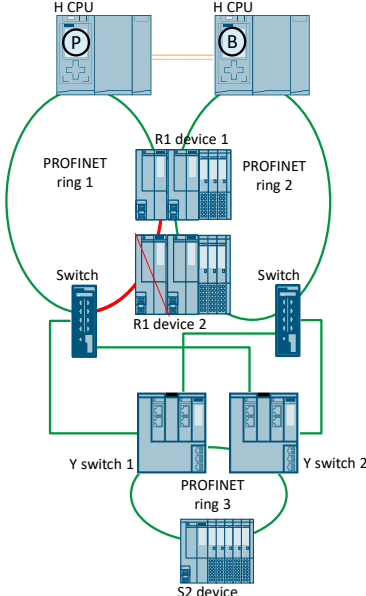
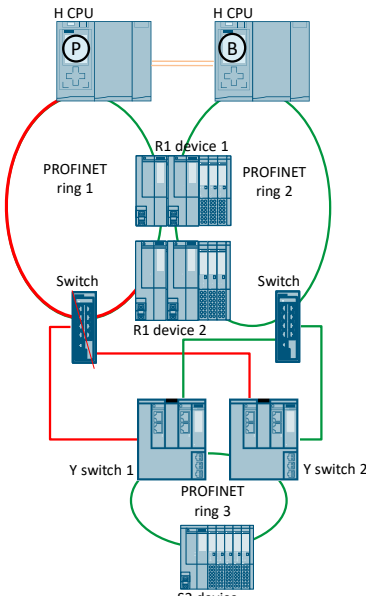
2 Configuration examples with R1 system redundancy

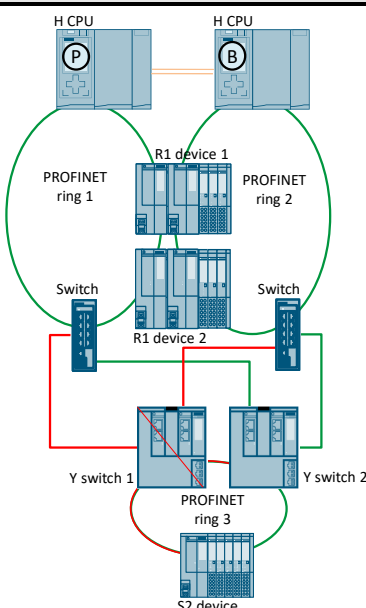
Scenario	Description
	The PROFINET line between the switch and the left-hand interface module of the "Y switch 1" device has failed. The primary CPU and backup CPU remain in the RUN Redundant mode. The roles of the primary and backup CPU do not change. The redundant system selects the path through the redundant Y switch "Y switch 2" as an alternative. The redundant system thereby reaches all IO devices in the PROFINET network again. The redundancy of the system is limited. This limitation has no impact on the process. If another system component or another network segment fails, then the redundant S7-1500H system can fail. The IO devices are not longer fully reachable in this case.
	The PROFINET line between the left-hand interface module of the Y switch "Y switch 1" and the device "S2 device" has failed. The roles of the primary and backup CPU do not change. The device "S2 device" remains reachable thanks to the ring topology. The redundancy of the system is limited. This limitation has no impact on the process. If another system component or another network segment fails, then the redundant S7-1500H system can fail. The IO devices are not longer fully reachable in this case.

Failure of one interface module

The Table shows possible redundancy scenarios should one interface module fail:

Table 2-9

Scenario	Description
	The left-hand interface module (slot 0) of the device "R1 device 2" fails. The roles of the primary and backup CPU do not change. The redundant system selects an alternative connection to the device "R1 device 2" via the backup CPU. The redundant system thereby reaches all IO devices in the PROFINET network again. The redundancy of the system is limited. This limitation has no impact on the process. If another system component or another network segment fails, then the redundant S7-1500H system can fail. The IO devices are not longer fully reachable in this case.
	The left switch fails. The roles of the primary and backup CPU do not change. The devices "R1 device 1" and "R1 device 2" remain reachable via the primary CPU. The redundant system selects an alternative connection to the Y switch via the backup CPU. The redundancy of the system is limited. This limitation has no impact on the process. If another system component or another network segment fails, then the redundant S7-1500H system can fail. The IO devices are not longer fully reachable in this case.

Scenario	Description
 The diagram illustrates a redundant S7-1500H system configuration. At the top, two H-CPU units are shown: 'H CPU P' (Primary) and 'H CPU B' (Backup). They are connected to two R1 devices, 'R1 device 1' and 'R1 device 2'. 'R1 device 1' is connected to 'PROFINET ring 1' and 'PROFINET ring 2'. 'R1 device 2' is connected to 'PROFINET ring 2' and 'PROFINET ring 3'. Two switches are connected to 'R1 device 1' and 'R1 device 2'. 'Y switch 1' is connected to 'R1 device 2' and 'PROFINET ring 3'. 'Y switch 2' is connected to 'R1 device 2' and 'PROFINET ring 3'. 'PROFINET ring 3' is connected to an 'S2 device'. A red 'X' is drawn over 'Y switch 1', indicating a failure. The diagram also shows 'PROFINET ring 1' and 'PROFINET ring 2' connected to the switches.	Both interface modules of the device "Y switch 1" have failed. The primary CPU and backup CPU remain in the RUN Redundant mode. The roles of the primary and backup CPU do not change. The devices "R1 device 1" and "R1 device 2" remain reachable via the primary CPU. The redundant system selects the path through the redundant Y switch "Y switch 2" as an alternative. The redundant system thereby reaches all IO devices in the PROFINET network again. The redundancy of the system is limited. This limitation has no impact on the process. If another system component or another network segment fails, then the redundant S7-1500H system can fail. The IO devices are not longer fully reachable in this case.

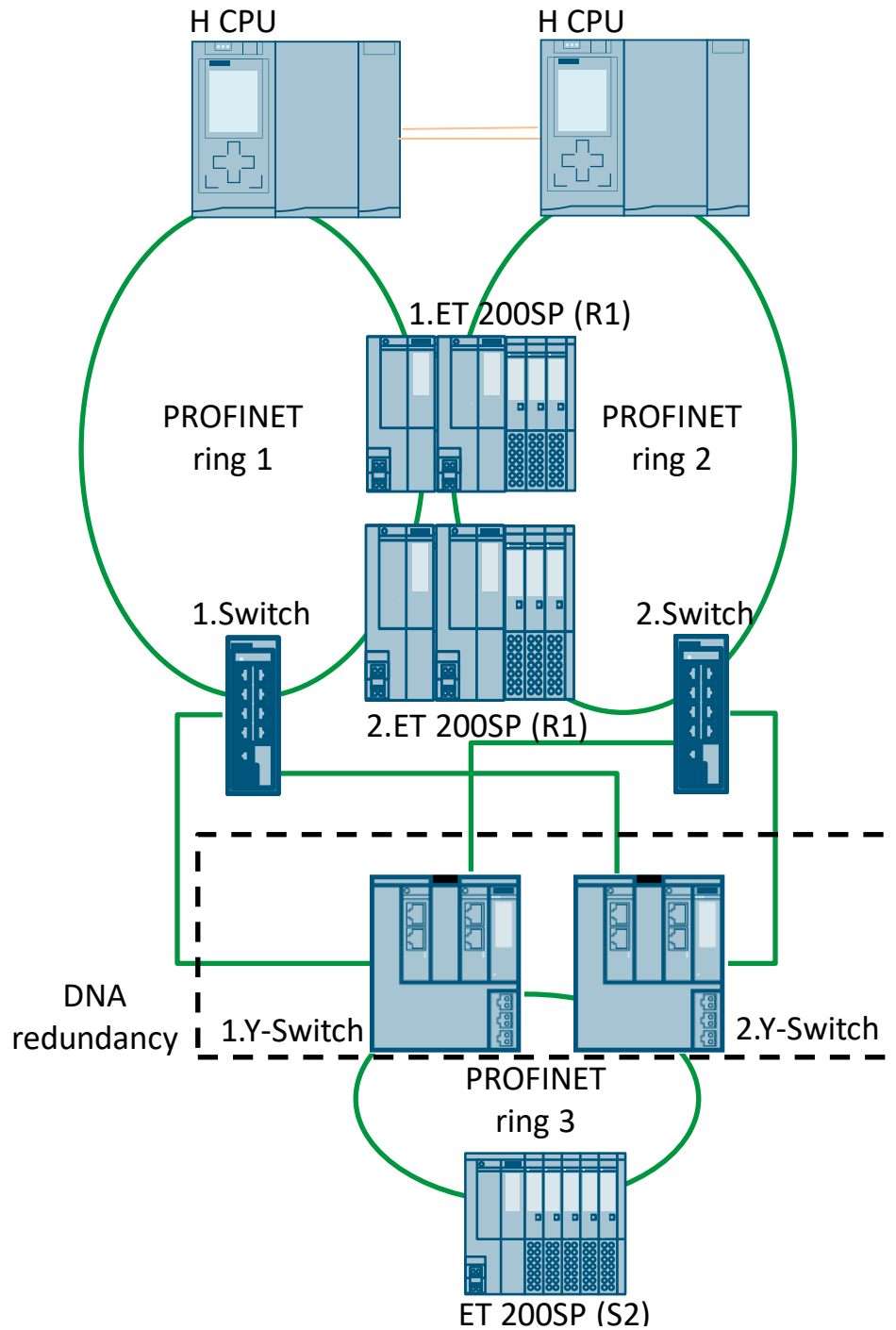
3 Engineering

3.1 Hardware configuration

Structure

The Figure below shows you the abstract structure of this application example:

Figure 3-1



IP addresses

The Table below gives you an overview of all IP addresses used in the example:

Table 3-1

Module	IP address
H-CPU (left) X1	192.168.0.1
H-CPU (right) X1	192.168.0.2
1.Switch	192.168.0.3
2.Switch	192.168.0.8
1.ET 200SP (R1) IM1	192.168.0.4
1.ET 200SP (R1) IM2	192.168.0.5
2.ET 200SP (R1) IM1	192.168.0.6
2.ET 200SP (R1) IM2	192.168.0.7
1.Y-Switch	192.168.0.9
2.Y-Switch	192.168.0.10
ET 200SP (S2)	192.168.0.11
PG	192.168.1.100

MRP role distribution

The Table below gives you an overview of the MRP roles and ring ports:

Table 3-2

Device	MRP domain	MRP role	Ring ports
H-CPU (left)	mrpdomain-1	Manager (Auto)	X1.P1 X1.P2
1.Switch	mrpdomain-1	Client	P1 P2
1.ET 200SP (R1) IM1	mrpdomain-1	Client	X1.P1 X1.P2
H-CPU (right)	mrpdomain-2	Manager (Auto)	X1.P1 X1.P2
2.Switch	mrpdomain-2	Client	P1 P2
2.ET 200SP (R1) IM2	mrpdomain-2	Client	X1.P1 X1.P2
1.Y-Switch	mrpdomain-3	Manager	P1.1 P2.1
2.Y-Switch	mrpdomain-3	Client	P1.1 P2.1
ET 200SP (S2)	mrpdomain-3	Client	X1.P1 X1.P2

DNA roles

The Table below gives you an overview of the DNA roles of the redundant Y switches:

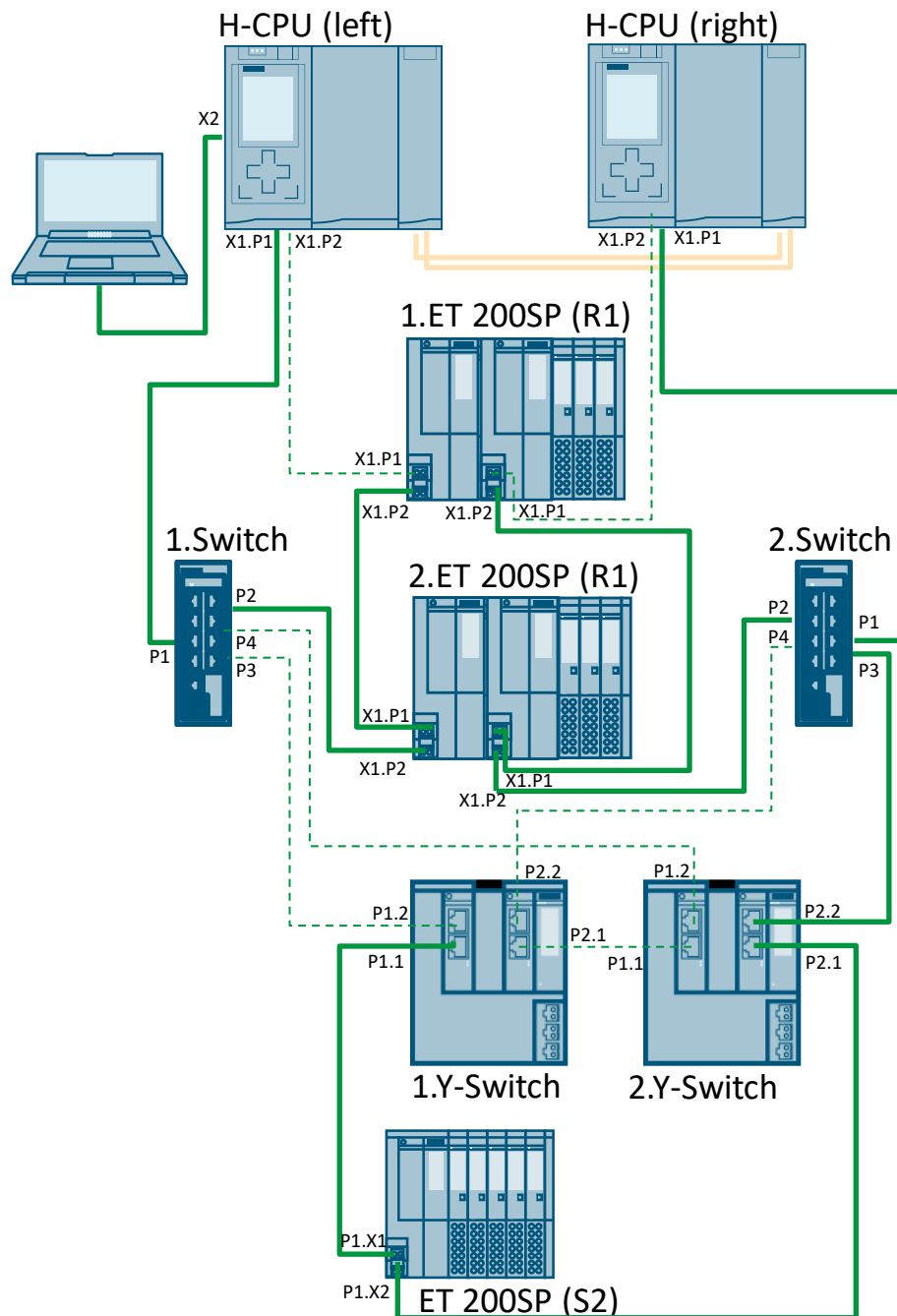
Table 3-3

Device	MRP role	DNA role	Ring ports	DNA ports
1.Y-Switch	Manager	DNA manager	P1.1 P2.1	P1.2 P2.2
2.Y-Switch	Client	DNA client	P1.1 P2.1	P1.2 P2.2

Topology

The following Figure shows the topology of the hardware setup.

Figure 3-2



ATTENTION

Cables represented with a dashed line may be connected with the partner device only after configuration is complete (after finishing [chapter 3.2](#)).

3.2 TIA Portal

Note

On the HTML page of this application example, you will find a preconfigured TIA Portal project for download.

If you are working with the included TIA Portal project, then [chapter 3.2](#) is solely for information purposes.

3.2.1 Hardware

Note

When configuring the hardware, it is assumed that your modules are at their factory settings.

Add hardware

Proceed as follows to create a new TIA Portal project with your hardware:

1. Open the TIA Portal V18.
2. Create a new project and give it a name.
3. In the hardware configuration, select the required modules from the hardware catalog in the Network view. For an overview of the modules used in this application example, see [chapter 1.3](#).
4. Drag the modules into the work pane of the Network view and assign them unique names.

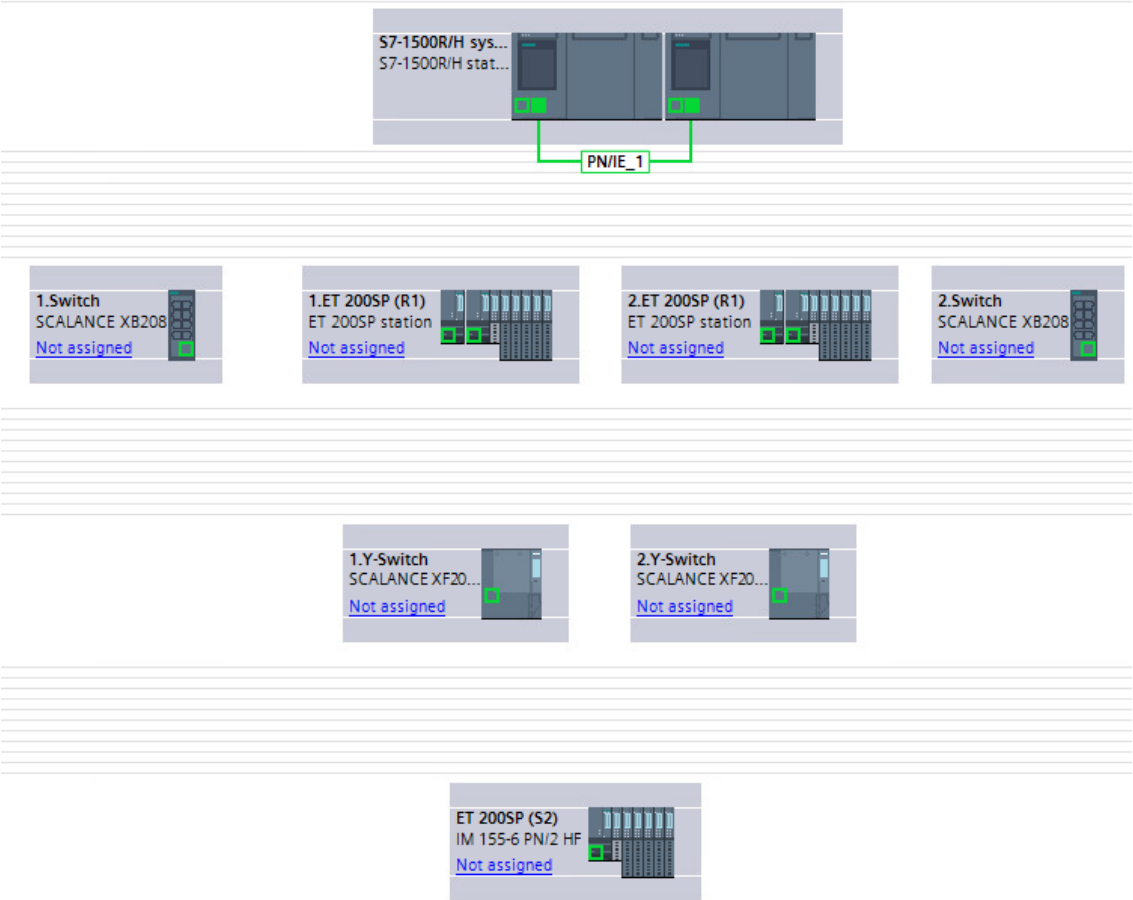
Note

When you drag the H CPU into your project, the Security Wizard will open automatically. Here, complete the security precautions for the H CPUs as your requirements dictate.

Result

You have created the modules. The PROFINET network between the H CPUs is created automatically.

Figure 3-3



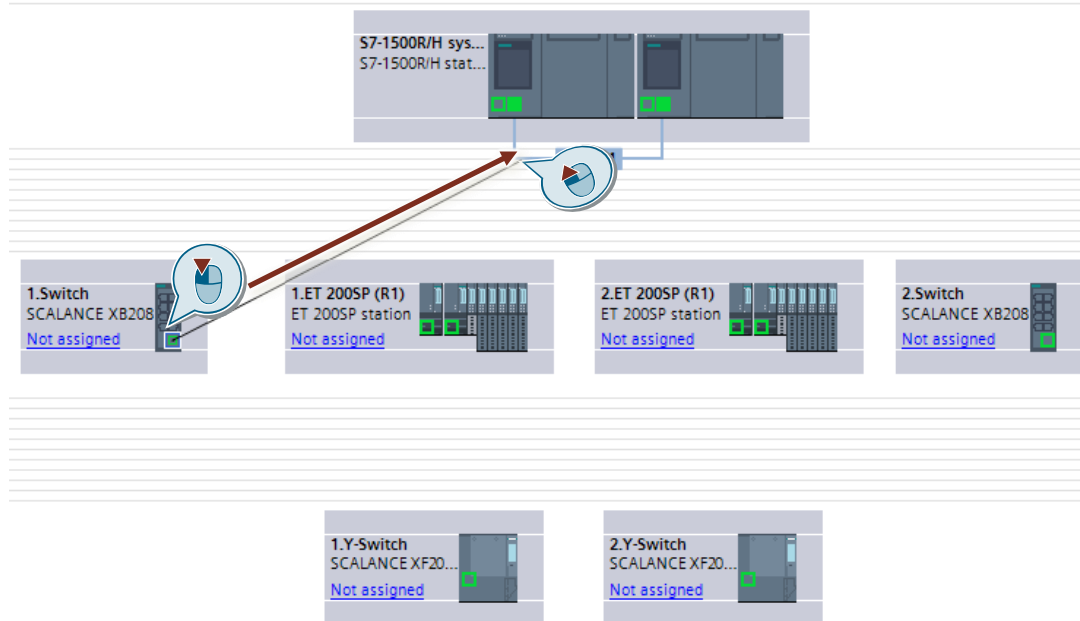
PROFINET subnet

The H CPUs and the IO devices are connected with a shared PROFINET subnet in TIA Portal. This configuration enables a two-way connection of S1 or S2 devices using a Y switch.

Proceed as follows to link the IO devices to the existing PROFINET subnet:

1. Highlight the PROFINET interface of an IO device and then drag and drop a line between the PROFINET interface of the IO device and the PROFINET network "PN/IE_1".

Figure 3-4



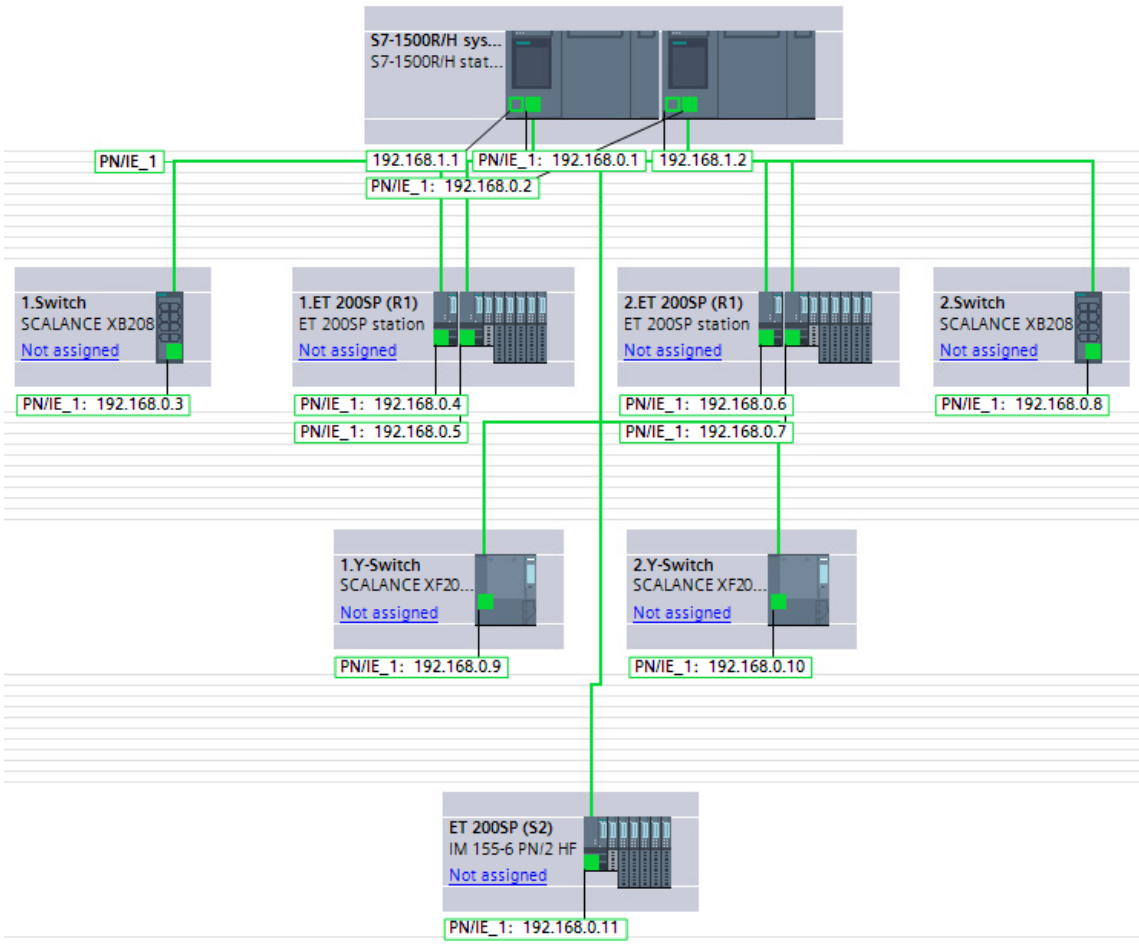
The IO device is connected with the PROFINET network. TIA Portal automatically assigns a consistent IP address. You can change this IP address if necessary.

2. Repeat Step 1 for all other IO devices.

Result

All IO devices are connected with the PROFINET network of the H CPUs and have an IP address.

Figure 3-5



Assign IO devices to the H CPUs

All IO devices must be assigned to the redundant system.

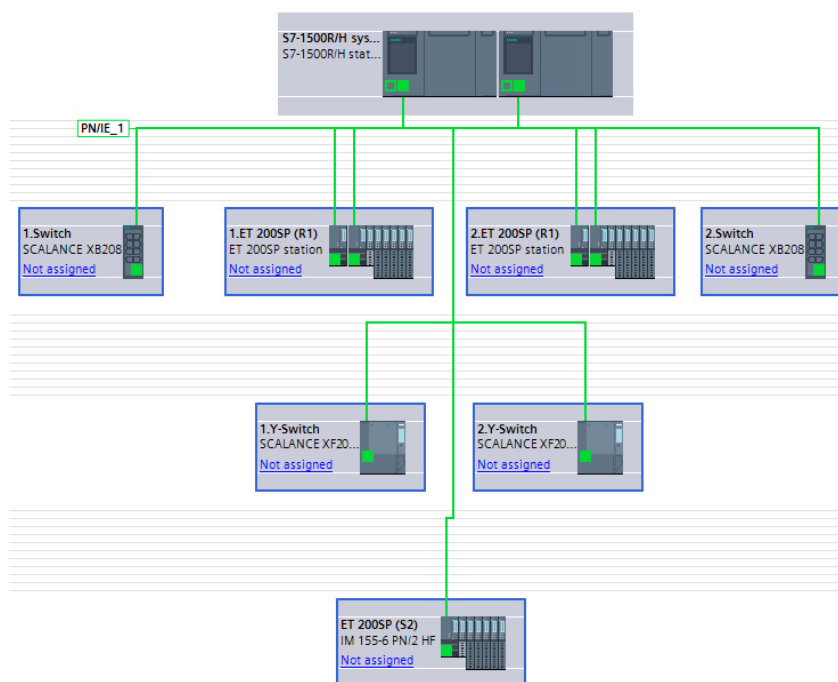
With IO devices with R1 system redundancy, connect each interface module of the R1 device with an H CPU in TIA Portal:

- Connect the left-hand interface module with the left H CPU in the Network view.
- Connect the right-hand interface module with the right H CPU in the Network view.

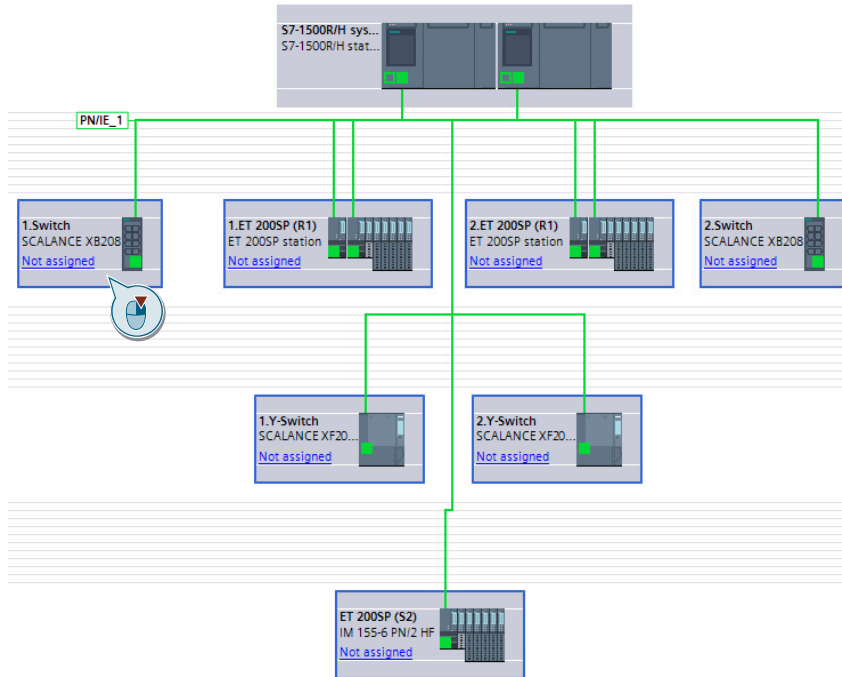
With IO devices with S2 system redundancy, connect the interface module of the IO device with every H CPU in the TIA Portal Network view.

To assign multiple IO devices to the redundant system at the same time, proceed as follows:

1. Highlight all IO devices in the Network view.

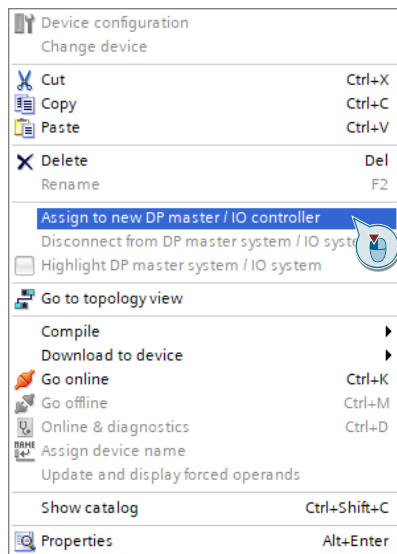


- Right-click an IO device to open the context menu.



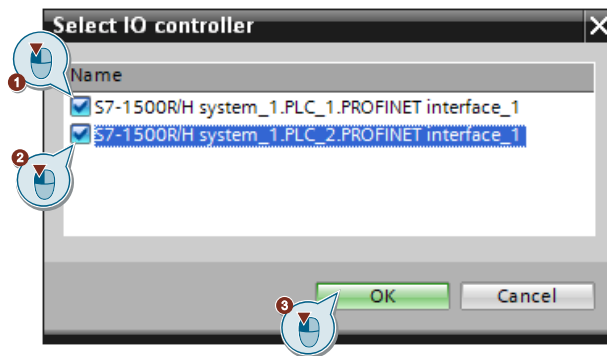
The context menu opens.

- Select "Assign to new DP master / IO Controller".



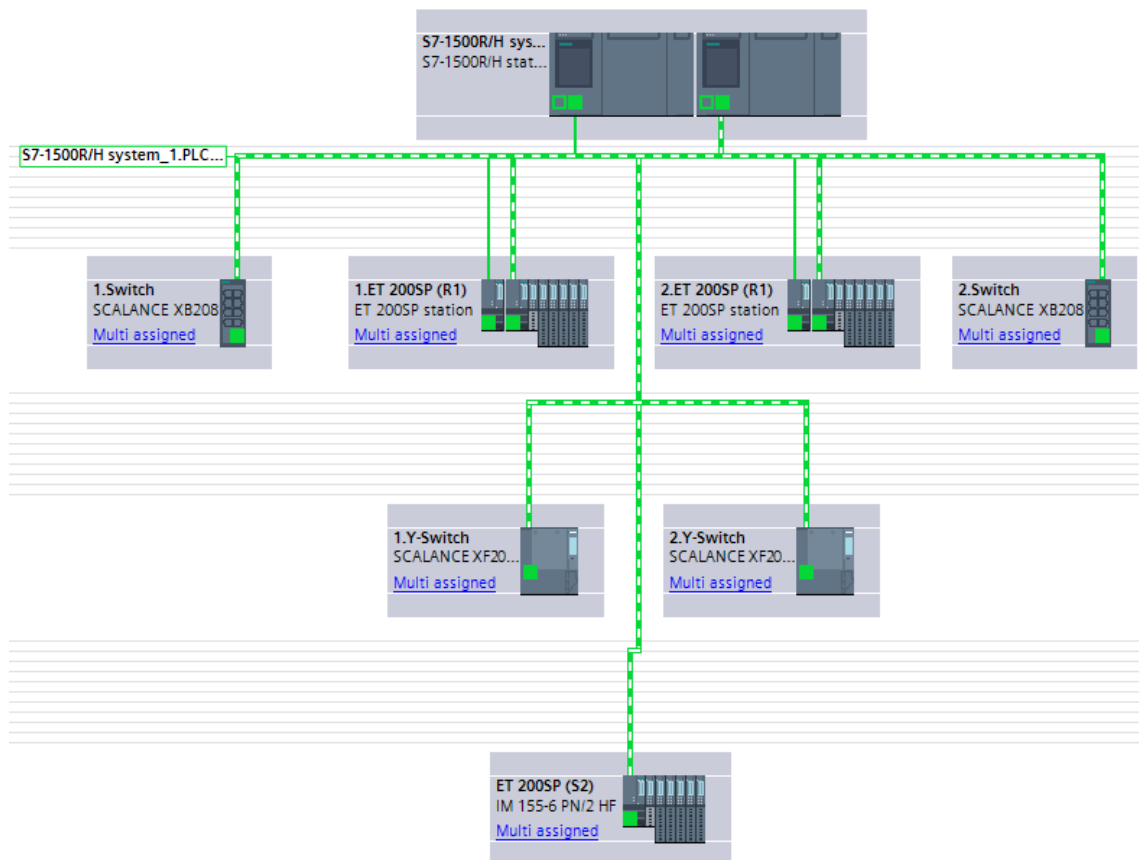
The "Select IO controller" dialog opens.

4. Select both H CPUs and click on "OK".



Result

All highlighted IO devices are assigned to the two H CPUs in accordance with the rules.



Reparameterize operating mode of the switches

Note

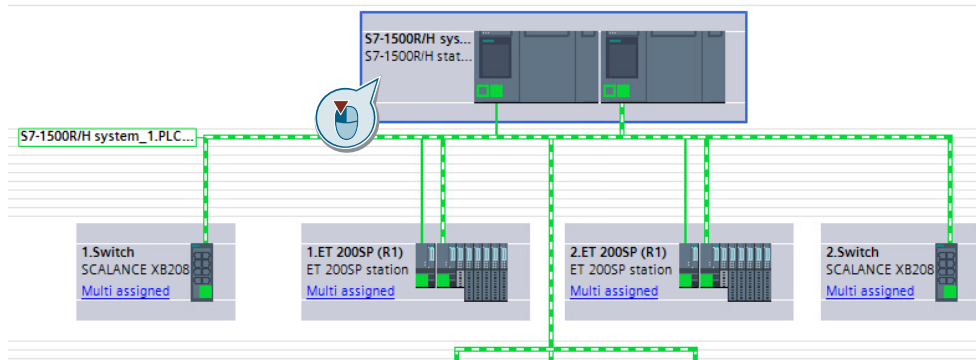
The SCALANCE XB208 switches used in the application example only support S1 system redundancy, meaning that in this case, you do not need to reparameterize the operating mode.

If you are using a switch with S2 system redundancy, then you must change the operating mode according to the instructions below.

If you are using a switch with S2 system redundancy, then the preset "IO device(S2)" operating mode will continually trigger the MAINT diagnostic signal on the H CPUs. To prevent this, change the operating mode of the two switches from "IO device(S2)" to "IO device(S1)".

Proceed as follows:

1. Highlight the redundant system in the Network view.



2. In the table view of the Network view, switch to "I/O communication".

Topology view Network view Device view				
Network overview Connections I/O communication VPN				
Device	Type	Address	Port	Subnet
▼ S7-1500R/H system_1	S7-1500R/H station			
▶ PLC_1	CPU 1517H-3 PN			
▶ PLC_2	CPU 1517H-3 PN			
▼ 1.ET 200SP (R1)	ET 200SP station			
▶ 1_ET 200SP (R1)_IM1	IM 155-6 PN R1			
▶ 1_ET 200SP (R1)_IM2	IM 155-6 PN R1			
▼ 2.ET 200SP (R1)	ET 200SP station			
▶ 2_ET 200SP (R1)_IM1	IM 155-6 PN R1			
▶ 2_ET 200SP (R1)_IM2	IM 155-6 PN R1			
▼ ET 200SP (S2)	ET 200SP station			
▶ ET 200SP (S2)	IM 155-6 PN/2 HF			
▼ 1.Y-Switch	SCALANCE XF204-2BA			
▶ 1.Y-Switch	SCALANCE XF204-2BA			
▼ SCALANCE XB-200	SCALANCE XB-200			
▶ 1.Switch	SCALANCE XB208			
▼ SCALANCE XB-200_1	SCALANCE XB-200			
▶ 2.Switch	SCALANCE XB208			
▼ 1.Y-Switch_1	SCALANCE XF204-2BA			
▶ 2.Y-Switch	SCALANCE XF204-2BA			

The "I/O communication" tab opens.

3. The "Mode" column shows you how the IO device is linked to the redundant S7-1500H system:

- IO device(S2): IO device is linked in redundant system.
- IO device(R1): IO device is linked in redundant system.
- IO device(S1): IO device is linked with the "Switched S1 device" function.

If you wish to change the mode of an IO device, then click the entry and select a different mode.

Network overview				
Connections				
I/O communication				
VPN				
Offline configuration				
Online assignment				
Partner 1	↔	Partner 2	Interface partner 2	Mode
1		PLC_1		
2		PROFI...		
3		X1	2.Y-Switch	SCALANCE interface_1
4		X1	2.Switch	SCALANCE interface_1
5		X1	1.Switch	SCALANCE interface_1
6		X1	1.Y-Switch	SCALANCE interface_1
7		X1	ET 200SP (S2)	PROFINET interface
8		X1	2_ET 200SP (R1)...	PROFINET interface
9		X1	1_ET 200SP (R1)...	PROFINET interface
10		PLC_2		
11		PROFI...		
12		X1	2.Y-Switch	SCALANCE interface_1
13		X1	2.Switch	SCALANCE interface_1
14		X1	1.Switch	SCALANCE interface_1
15		X1	1.Y-Switch	SCALANCE interface_1
16		X1	ET 200SP (S2)	PROFINET interface
17		X1	2_ET 200SP (R1)...	PROFINET interface

Result

The mode of the IO device has been modified.

3.2.2 PROFINET ring

Overview

In this application example, communication runs on multiple separate PROFINET rings (MRP domains).

- Two PROFINET rings for R1 device communication
- One PROFINET ring in the lower-level network due to DNA redundancy

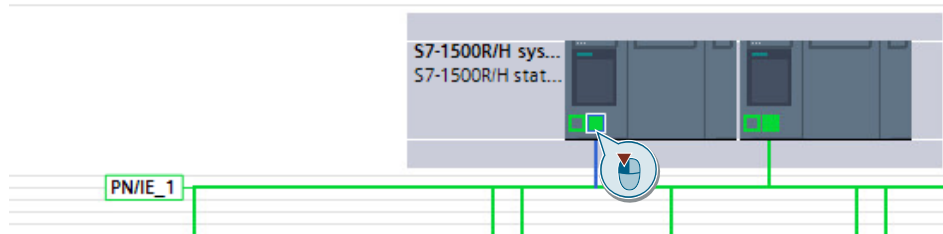
Note

TIA Portal V18 already has four MRP domains created by default at switches. You do not need to create any new MRP domains.

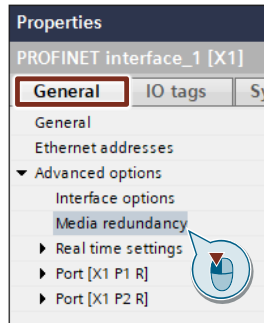
MRP role and MRP domain of the H CPUs

Proceed as follows to define the MRP role and MRP domain of the H CPUs:

1. In the Network view, select the "X1" PROFINET interface of the left-hand H CPU.

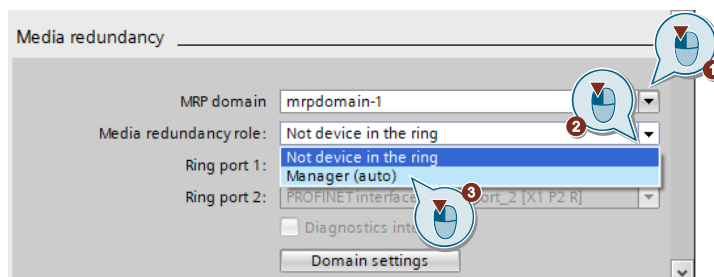


2. In the Inspector window, navigate to "Properties > General > Advanced options > Media redundancy".



The "Media redundancy" section opens.

3. Select the MRP domain and the MRP role according to [Table 3-2](#).



4. In the Network view, select the "X1" PROFINET interface of the right-hand H CPU, and repeat Steps 2 and 3.

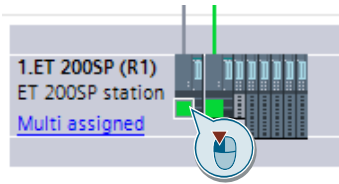
Result

The H CPUs are assigned to their MRP domain as MRP manager (auto).

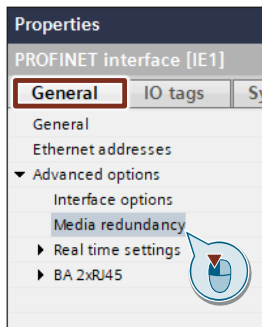
MRP role and MRP domain of the R1 devices

Proceed as follows to define the MRP role and MRP domain of the R1 devices:

1. In the Network view, select the PROFINET interface of the "IM1" interface module for the device "1.ET 200SP (R1)".

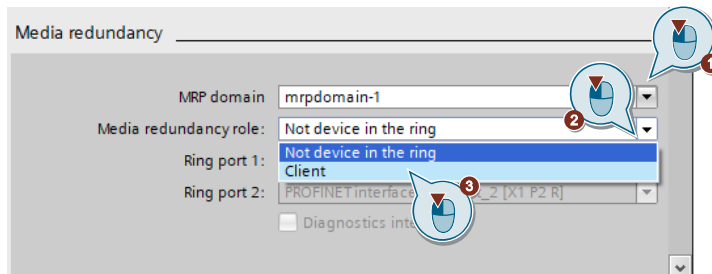


2. In the Inspector window, navigate to "Properties > General > Advanced options > Media redundancy".



The "Media redundancy" section opens.

3. Select the MRP domain and the MRP role according to [Table 3-2](#).



4. In the Network view, select the PROFINET interface of the "IM2" interface module for the device "1.ET 200SP (R1)" and repeat Steps 2 and 3.
5. Repeat steps 1-4 for the device "2.ET 200SP (R1)".

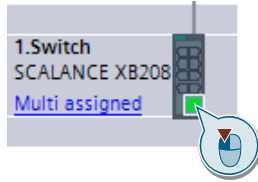
Result

The interface modules of the two ET 200SP (R1) devices have been assigned to their MRP domain as MRP clients.

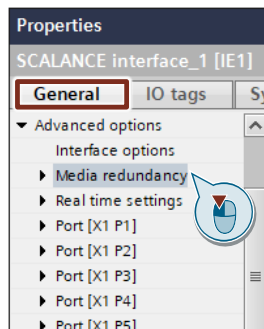
MRP role and MRP domain of the switches

To define the MRP roles, MRP domain and the ring ports of the two switches, proceed as follows:

1. In the Network view, select the PROFINET interface of the device "1.Switch".

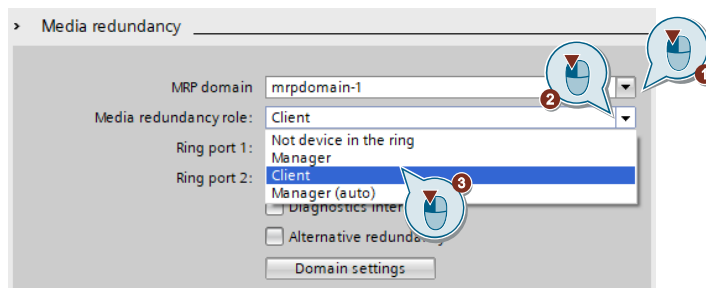


2. In the Inspector window, navigate to "Properties > General > Advanced options > Media redundancy".

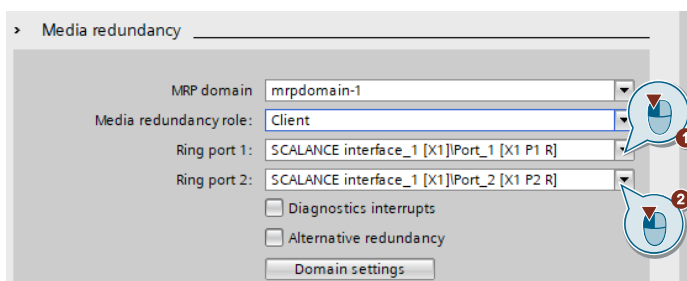


The "Media redundancy" section opens.

3. In the "Media redundancy" section, select the MRP domain and the MRP role as specified in [Table 3-2](#).



4. Select the ring ports according to [Table 3-2](#).



5. In the Network view, select the PROFINET interface of the device "2.Switch", and repeat Steps 2 through 4.

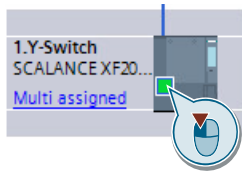
Result

The two switches have been assigned to their MRP domain as MRP clients.

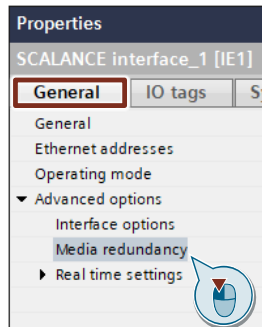
MRP role and MRP domain of the Y switches

To define the MRP roles, MRP domain and the ring ports of the Y switches, proceed as follows:

1. In the Network view, select the PROFINET interface of the device "1.Y-Switch".

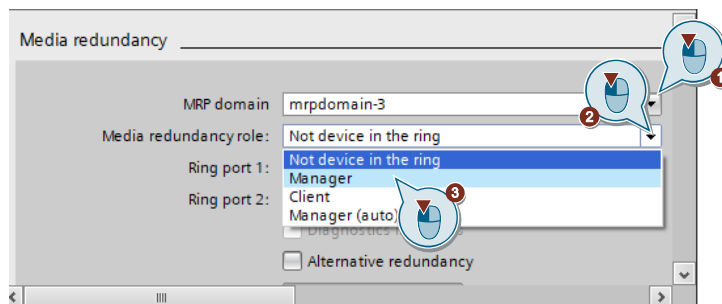


2. In the Inspector window, navigate to "Properties > General > Advanced options > Media redundancy".

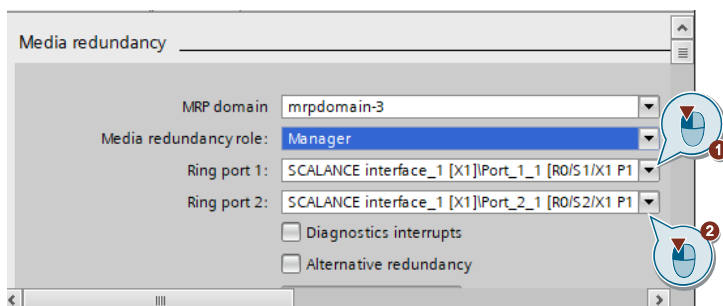


The "Media redundancy" section opens.

3. Select the MRP domain and the MRP role according to [Table 3-2](#).



4. Select the ring ports according to [Table 3-2](#).



5. In the Network view, select the PROFINET interface of the device "2.Y-Switch", and repeat Steps 2 through 4.

Result

The two Y switches have been assigned to their MRP domain as MRP manager or MRP client.

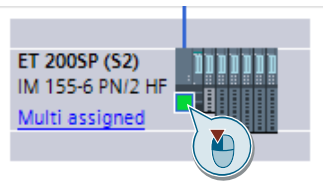
Note

Ports not configured as MRP ring ports are automatically DNA ports.

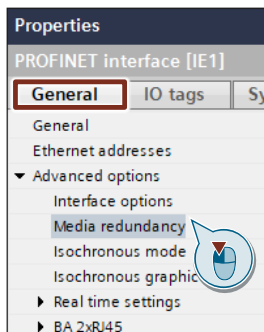
MRP role and MRP domain of the ET 200SP (S2)

Proceed as follows to define the MRP roles and MRP domain of the ET 200SP (S2):

1. In the Network view, select the PROFINET interface of the device "ET 200SP (S2)".

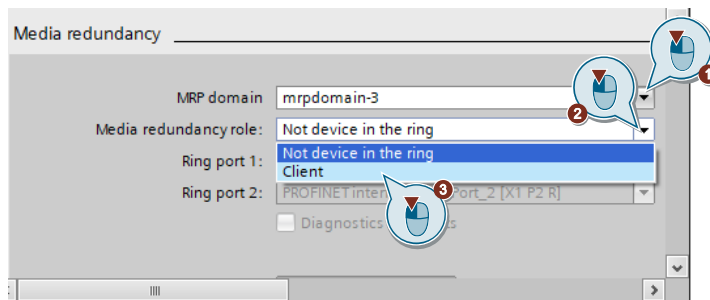


2. In the Inspector window, navigate to "Properties > General > Advanced options > Media redundancy".



The "Media redundancy" section opens.

3. Select the MRP domain and the MRP role according to [Table 3-2](#).



Result

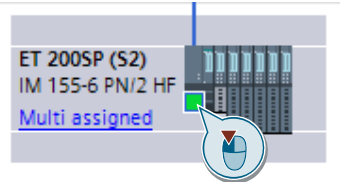
The device "ET 200SP (S2)" has been assigned to its MRP domain as an MRP client.

Response watchdog time

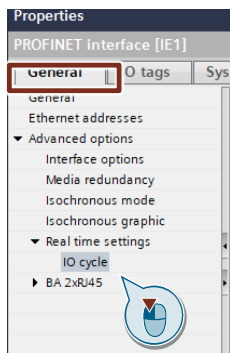
The reconfiguration time of the rings is typically less than 200 milliseconds. In order for the data exchange to work after a reconfiguration, you must set the response watchdog time of the IO devices that are nodes in a PROFINET ring to be greater than 200 milliseconds.

To inspect and increase the response watchdog time of the IO devices, proceed as follows:

1. In the Network view, select the PROFINET interface of the device "ET 200SP (S2)".

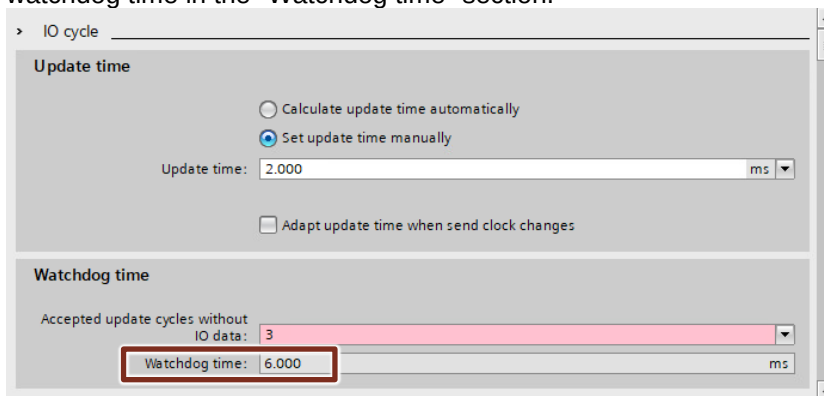


2. In the Inspector window, navigate to "Properties > General > Advanced options > Real time settings > IO cycle".



The "IO cycle" section opens.

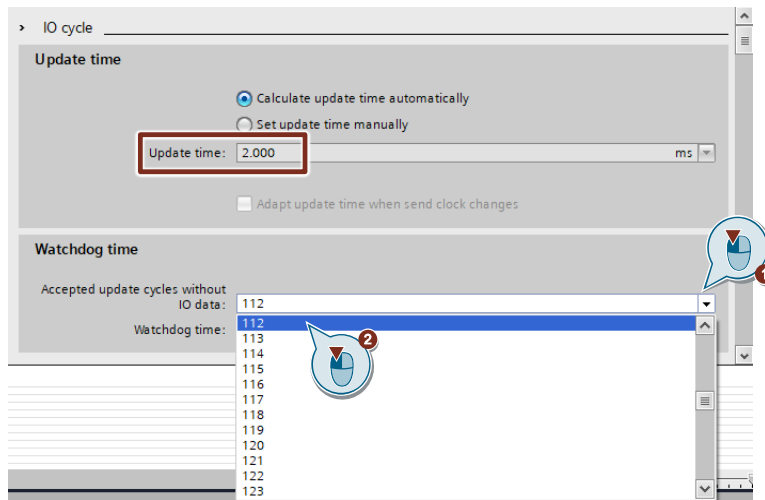
3. If the IO device is a node in a PROFINET ring, the response watchdog time must be greater than the reconfiguration time of the ring. You can check the currently configured response watchdog time in the "Watchdog time" section.



Note

TIA Portal will highlight the parameter "Accepted update cycles without IO data" in red if the response watchdog time is too low. Follow Step 4 to adjust the response watchdog time. If the parameter is not highlighted, you do not need to make any further changes.

4. The response watchdog time is a whole-number multiple of the "Update time" (here: 2,000 milliseconds). To increase the response watchdog time, select a suitable factor for "Accepted update cycles without IO data" (here: factor 112). The following calculation is used: Response watchdog time = update time * factor.

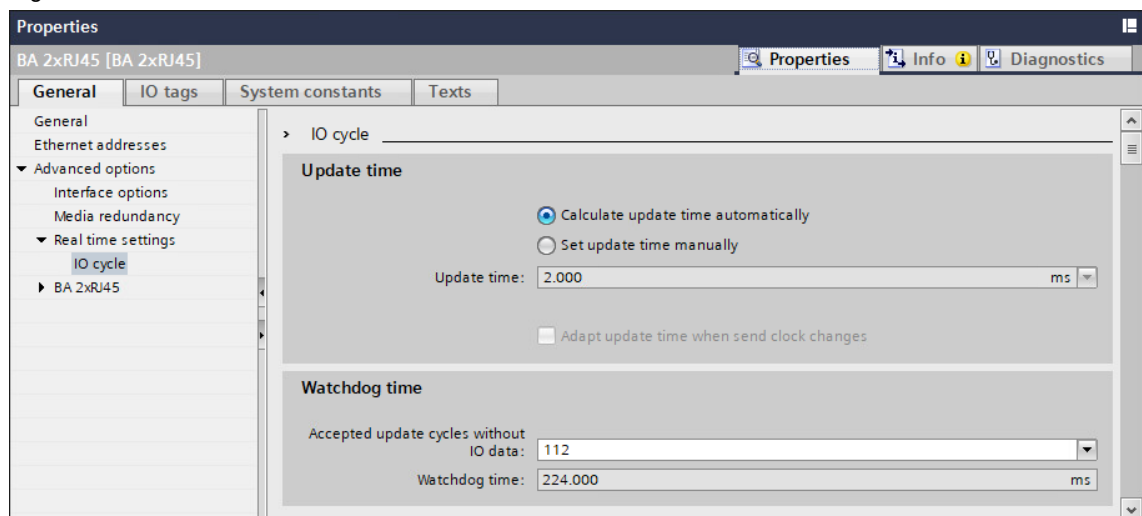


5. Repeat Step 1-4 for all other IO devices that are nodes in a PROFINET ring.

Result

The response watchdog time of the IO devices in the PROFINET ring has been adjusted, and the time settings are now higher than the reconfiguration time of the PROFINET ring.

Figure 3-6

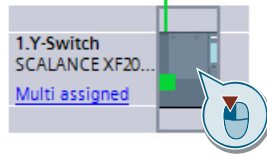


3.2.3 DNA redundancy

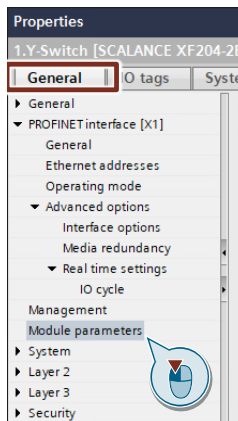
The Y switch in this application example has a redundant design.

To enable DNA redundancy in the Y switches, proceed as follows:

1. In the Network view, select the device "1.Y-Switch".

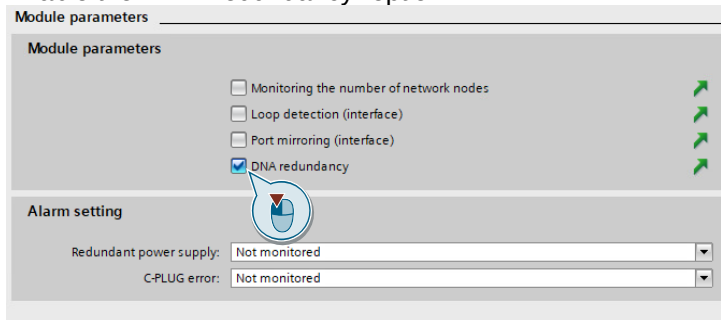


2. In the Inspector window, navigate to "Properties > General > PROFINET interface[X1] > Module parameters".

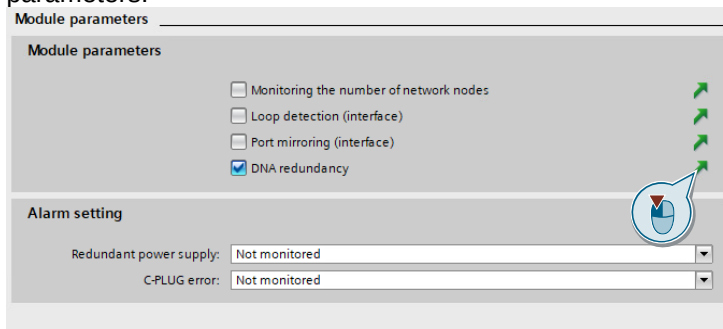


The "Module parameters" section opens.

3. Enable the "DNA redundancy" option.

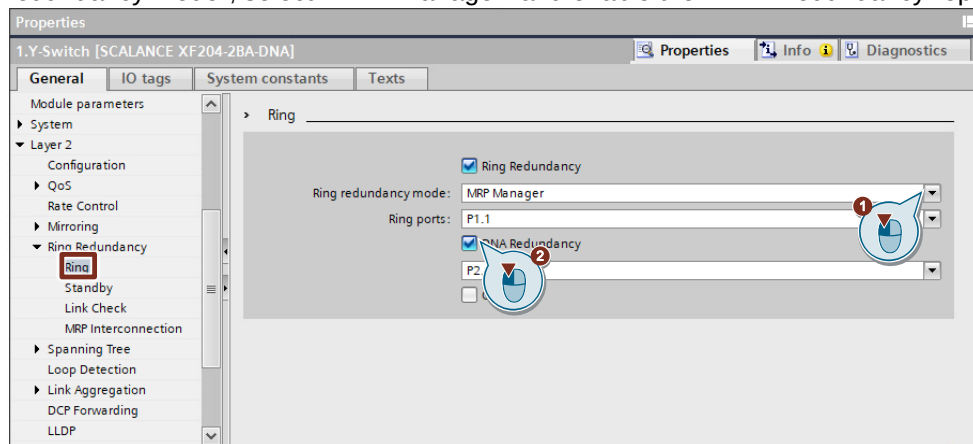


- Additional parameters must be adjusted. Click the green arrow to go to the other parameters.



TIA Portal will automatically navigate you to the menu "Properties > General > Layer 2 > Ring Redundancy > Ring".

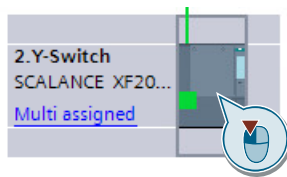
- The device "1.Y-Switch" assumes the MRP manager role (see [chapter 3.2.2](#)). For "Ring redundancy mode", select "MRP Manager" and enable the "DNA Redundancy" option.



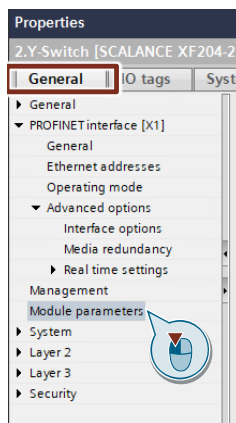
Result

The device "1.Y-Switch" has been configured as MRP manager, and simultaneously assumes the role of "DNA manager".

6. In the Network view, select the device "2.Y-Switch".

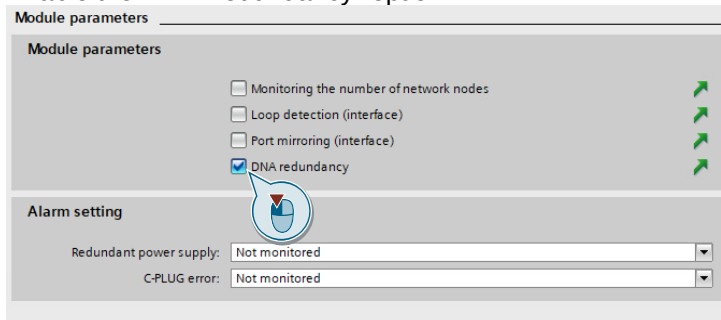


7. In the Inspector window, navigate to "Properties > General > PROFINET interface[X1] > Module parameters".

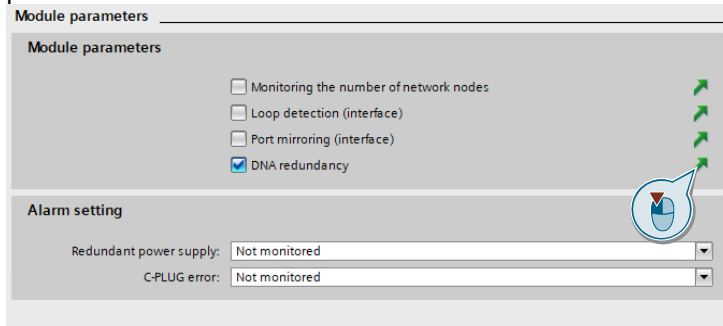


The "Module parameters" section opens.

8. Enable the "DNA redundancy" option.

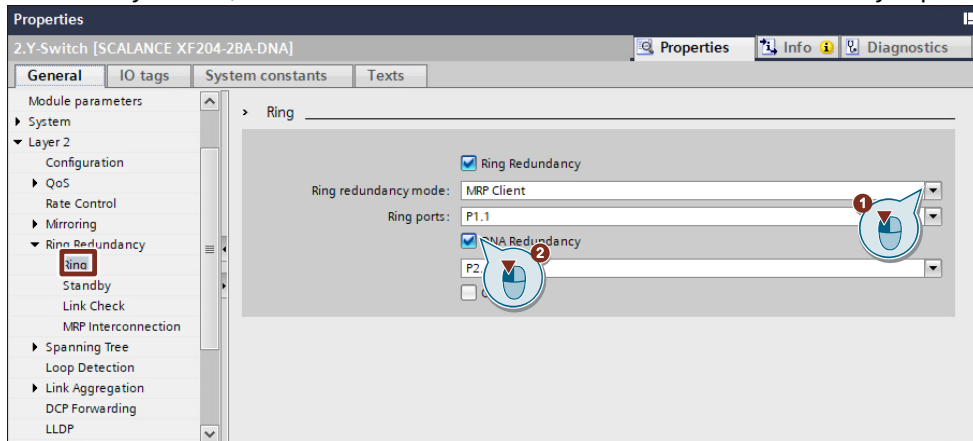


9. Additional parameters must be adjusted. Click the green arrow to go to the other parameters.



TIA Portal will automatically navigate you to the menu "Properties > General > Layer 2 > Ring Redundancy > Ring".

10. The device "2.Y-Switch" assumes the MRP client role (see [chapter 3.2.2](#)). For "Ring redundancy mode", select "MRP Client" and enable the "DNA Redundancy" option.



Result

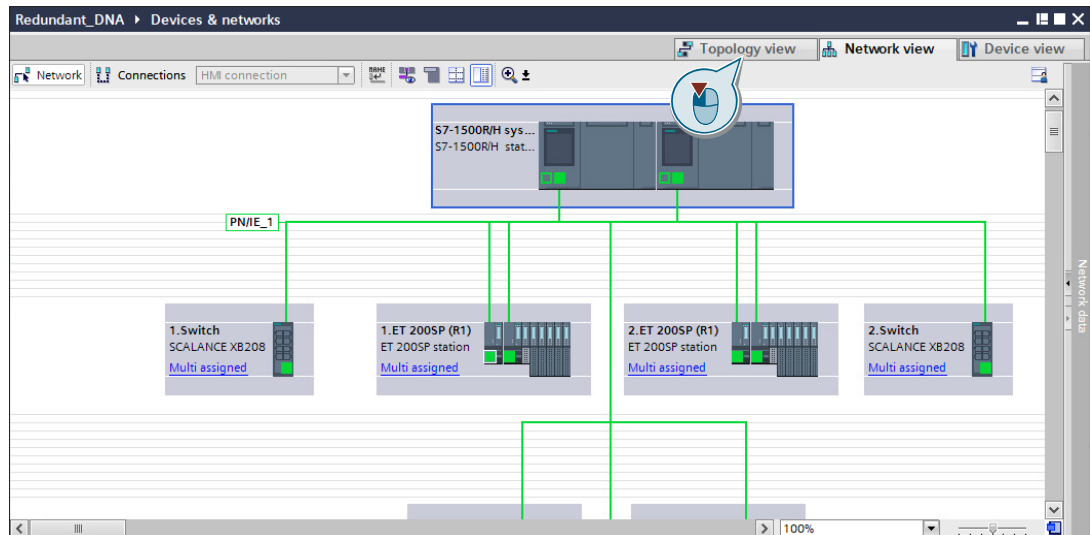
The device "2.Y-Switch" has been configured as MRP client, and simultaneously assumes the role of "DNA client".

3.2.4 Topology

When configure the TIA project as a topology, then TIA Portal will check the consistency. An invalid configuration will be indicated by error messages.

Proceed as follows to configure the topology:

1. Switch to the "Topology view" tab.



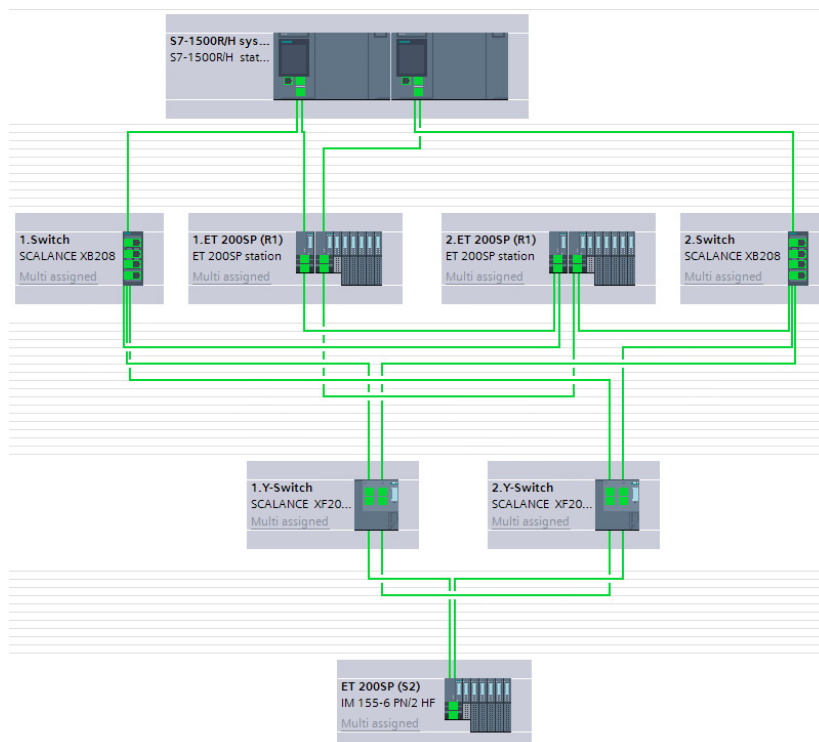
The project opens in the Topology view.

2. Drag and drop to connect the ports with each other according to [Figure 3-2](#).

Result

The topology is configured.

Figure 3-7



3.3 Final steps

3.3.1 Initial commissioning and download

Prerequisite

All modules are connected to a power supply.

Initial commissioning

During the initial configuration, all IO devices require a PROFINET device name.

1. Connect the programming device with each interface module of the IO devices.
2. Open the TIA Portal project, included here or created in [chapter 3.2](#).
3. Give the modules the PROFINET device name generated in TIA Portal. Use the functions in TIA Portal for this.

Download project

Proceed as follows to download the project to the H CPU:

1. Connect the programming device with the [X2] interface of the H CPU (primary).
2. Download the TIA Portal project to the H CPU.

3.3.2 Physical connection

Connect the devices

Connect all IO devices and H CPUs as shown in [Figure 3-2](#).

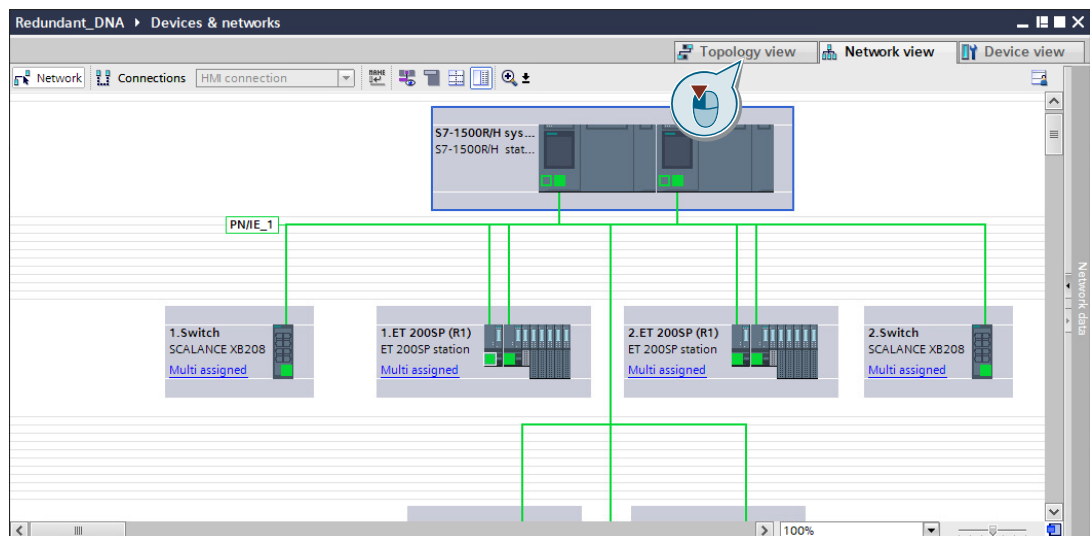
Note

Since the H CPUs have the current configuration, you can close the connections that are shown as open in the Figure.

Check topology

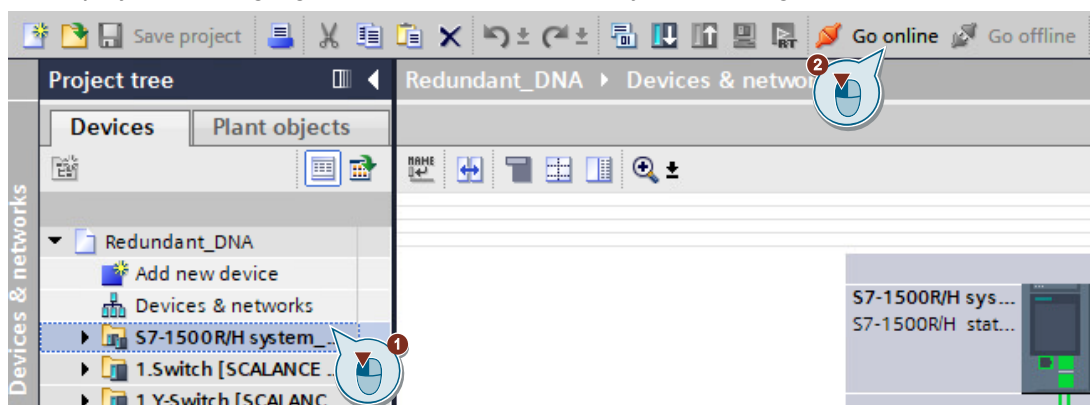
To check the topology live, proceed as follows:

1. Switch to the "Topology view" tab in TIA Portal.



The project opens in the Topology view.

2. In the project tree, highlight the redundant S7-1500 system, then go online.

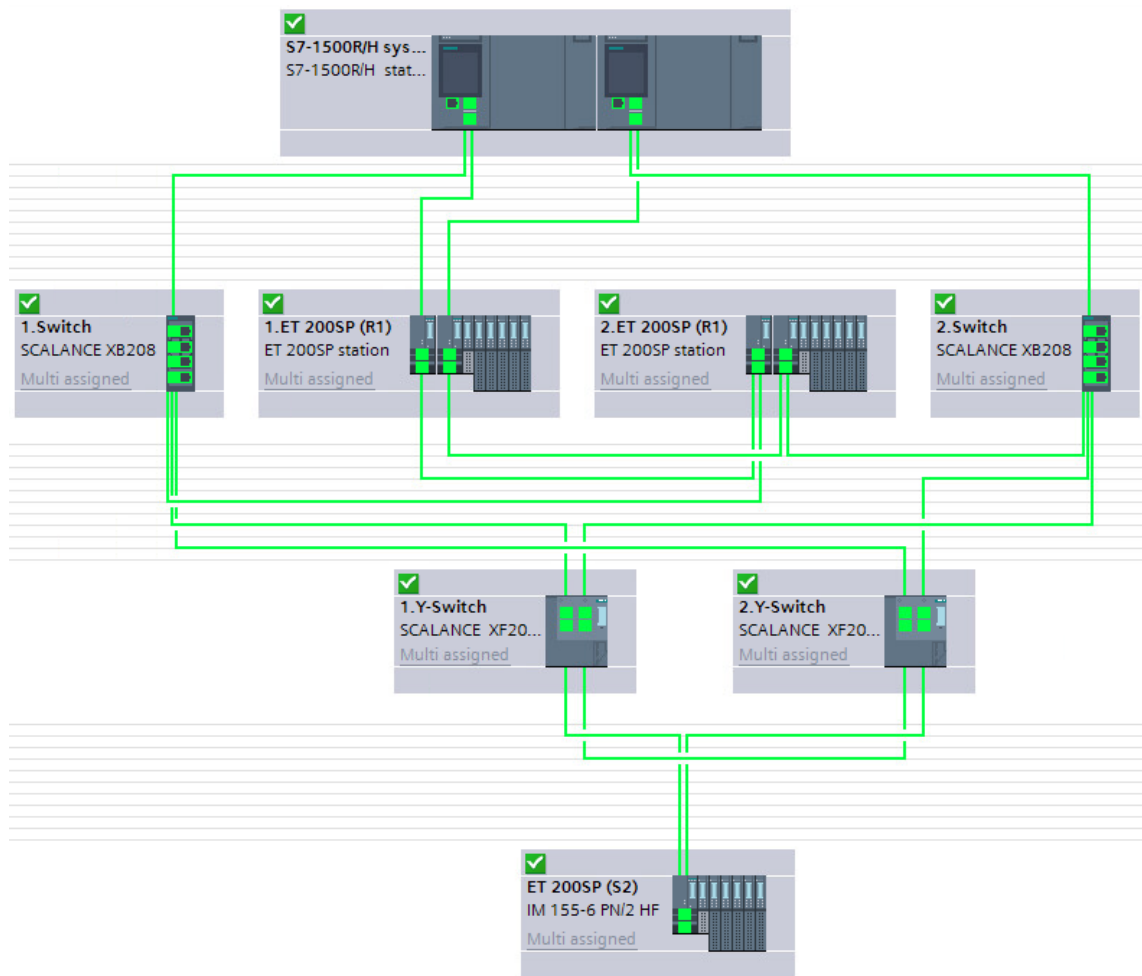


The Topology view opens in the online view.

Result

You can see the status of the devices and connections in the Topology view.

Figure 3-8



Note

If connections, ports or devices are faulty or invalid, TIA Portal will highlight the corresponding object in color.

Remedy the errors.

3.4 Operation

The application example allows for uninterrupted process data exchange in the event of:

- Failure of an H CPU
- Interruption of the PROFINET ring
- Failure of one interface module
- Failure of a subnet

Failure of an H CPU

Remove the supply voltage of the H CPU (primary), thereby simulating the failure of the H CPU. The redundant system switches over to the backup CPU. The backup CPU becomes the new primary CPU and switches to the system state RUN Solo. All IO devices remain reachable.

Reconnect the H CPU to the power supply.

Interruption of the PROFINET ring

To simulate the interruption of PROFINET ring 1, remove the PROFINET cable, for example, the one between the primary CPU and the switch.

The primary CPU and backup CPU remain in the RUN Redundant mode. All IO devices remain reachable.

Reconnect the primary CPU and the switch.

Failure of an interface module

To simulate the failure of an interface module, remove the power supply for the left-hand interface module (slot 0) of the device "2.ET 200SP (R1)".

The roles of the primary and backup CPU do not change. The redundant system selects an alternative connection to the device "2.ET 200SP (R1)" via the backup CPU. The redundant system thereby reaches all IO devices in the PROFINET network again.

Reconnect the interface module to the power supply.

Failure of a subnet

To simulate the failure of PROFINET ring 3, remove the PROFINET cable, for example, the one between the device "1.Y-Switch" and the device "ET 200SP (S2)".

The primary CPU and backup CPU remain in the RUN Redundant mode. The device "ET 200SP (S2)" remains reachable thanks to the ring topology.

Reconnect the device "1.Y-Switch" and the device "ET 200SP (S2)".

4 Useful information

4.1 Explanation of redundancy terminology

System redundancy terminology

PROFINET system redundancy distinguishes between four configurations, known as S1, S2, R1 and R2.

The shorthand "S" stands for "singular" and describes an IO device with one interface module. The shorthand "R" stands for "redundant". The following variations of an IO device are possible:

- Two interface modules, each with one PN interface.
- One interface module with two independent PN interfaces.

An "S" device is therefore able to connect with a singular network. By contrast, an "R" device can connect to a redundant network.

The number after the letter signifies the maximum number of Application Relationships (abbreviated as AR) for an interface module.

With S1 and R1 devices, there is exactly one connection to the CPU from the interface module.

S1 devices are typically run connected with a non-redundant controller. However, they can also be used in the S7-1500H system with some restrictions.

R1 devices, on the other hand, require a fault-tolerant system to run. This variant involves establishing one connection between each redundant interface module and redundant CPU (two connections in total).

S2 devices are able to establish two connections. This enables them to be run on a redundant system without limitations, even though there is only one interface module in the device.

R2 devices would each establish two connections to the fault-tolerant system (that is, 4 connections in total). This operating mode is not implemented in SIMATIC products, however, since the negligible increase in availability does not justify the high level of system complexity that is necessary.

Note

In Siemens Industry Online Support, you can find a FAQ that describes the redundancy terms in more detail. The link to the FAQ can be found in reference \3\ in [chapter 5.3](#).

Connection to redundant system

You can use the following IO devices on the redundant S7-1500H system:

- IO devices with S2 system redundancy
- IO devices with R1 system redundancy (only S7-1500H with firmware version V3.0 or later)
- Standard IO devices, via the "Switched S1 device" function of the H CPU

Compensation in case of a fault

IO devices with S2 system redundancy allow for uninterrupted process data exchange with the redundant S7-1500H system in the event of:

- Failure of an H CPU
- Interruption of the PROFINET ring
- Interruption of the linear topology (with S7-1500H as of firmware version V3.0 or later)

IO devices with R1 system redundancy allow for uninterrupted process data exchange with the redundant S7-1500H system in the event of:

- Failure of an H CPU
- Interruption of the PROFINET ring or linear topology
- Failure of one interface module
- Failure of a subnet

A switched S1 device does not tolerate an interruption of IO controller or IO device communication. The S1 device becomes temporarily unavailable, and then it may be re-acquired.

4.2 Requirements

Note the following requirements when using the redundant S7-1500H system:

Table 4-1

Category	Description
CPU	You need two identical H CPUs (item number and firmware version) in the redundant system.
MRP	- All PROFINET devices in the PROFINET ring support the Media Redundancy function. - The two H CPUs have the media redundancy role "Manager (auto)". All other nodes in the PROFINET ring have the media redundancy role "Client".
Redundancy	- Using the "Switched S1 device" function, you can run any standard IO device on the redundant S7-1500H system. - For uninterrupted exchange of process data in the event that the primary CPU fails (primary-backup switchover), an IO device must support S2 or R1 system redundancy.

5 Appendix

5.1 Service and support

Industry Online Support

Do you have any questions or need assistance?

Siemens Industry Online Support offers round the clock access to our entire service and support know-how and portfolio.

The Industry Online Support is the central address for information about our products, solutions and services.

Product information, manuals, downloads, FAQs, application examples and videos – all information is accessible with just a few mouse clicks:

support.industry.siemens.com

Technical Support

The Technical Support of Siemens Industry provides you fast and competent support regarding all technical queries with numerous tailor-made offers

– ranging from basic support to individual support contracts. Please send queries to Technical Support via Web form:

siemens.com/SupportRequest

SITRAIN – Digital Industry Academy

We support you with our globally available training courses for industry with practical experience, innovative learning methods and a concept that's tailored to the customer's specific needs.

For more information on our offered trainings and courses, as well as their locations and dates, refer to our web page:

siemens.com/sitrain

Service offer

Our range of services includes the following:

- Plant data services
- Spare parts services
- Repair services
- On-site and maintenance services
- Retrofitting and modernization services
- Service programs and contracts

You can find detailed information on our range of services in the service catalog web page:

support.industry.siemens.com/cs/sc

Industry Online Support app

You will receive optimum support wherever you are with the "Siemens Industry Online Support" app. The app is available for iOS and Android:

support.industry.siemens.com/cs/www/en/sc/2067

5.2 Industry Mall



The Siemens Industry Mall is the platform on which the entire Siemens Industry product portfolio is accessible. From the selection of products to the order and the delivery tracking, the Industry Mall enables the complete purchasing processing – directly and independently of time and location:

mall.industry.siemens.com

5.3 Links and literature

Table 5-1

No.	Topic
\1\	Siemens Industry Online Support https://support.industry.siemens.com
\2\	Link to the article page of the application example https://support.industry.siemens.com/cs/ww/en/view/109816704
\3\	FAQ: What are the PROFINET redundancy functions? https://support.industry.siemens.com/cs/ww/en/view/109756450
\4\	Configuration manual: SCALANCE XB-200/XC-200/XF-200BA/XP-200/XR-300WG Web Based Management https://support.industry.siemens.com/cs/ww/en/view/109780061
\5\	Manual: SIMATIC S7-1500 S7-1500R/H redundant system https://support.industry.siemens.com/cs/de/en/view/109754833
\6\	Manual: SIMATIC PROFINET PROFINET with STEP 7 https://support.industry.siemens.com/cs/ww/en/view/49948856
\7\	System manual: SIMATIC ET 200SP Distributed I/O system https://support.industry.siemens.com/cs/de/en/view/58649293
\8\	S7-1500R/H add-in https://support.industry.siemens.com/cs/ww/en/view/109769093

5.4 Change documentation

Table 5-2

Version	Date	Change
V1.0	03/2023	First edition